Compute Cell Scheduling Records

This document records how tasks were scheduled on small compute clusters. The workload and the machine failures are real records from a production compute cell of about 12,500 machines (the Google cluster trace of May 2011, clusterdata-2011-2, CC-BY 4.0): every task's submission time, job, priority, CPU and memory request and run time, and every machine's capacity, failure time and return time, are taken from that trace (ids are renamed). Where and when each task ran in the small cluster is not recorded in the trace: it is computed from those records by the scheduling policy stated below, applied exactly.

The document is divided into chapters titled 'Cell window <NN>'. Each chapter is an independent run of one small cluster over a window of 1200 seconds. Questions refer to a chapter by that title. Times are whole seconds from the start of the window (time 0); the window covers times 0 to 1200.

Each chapter has four sections. 'Machines' lists the cluster's machines with their CPU and memory capacity, the time a machine failed (if it did) and the time it returned (if it did). 'Task log' lists, in time order, every event of every task from time 0 to the window end: submit, start, finish, evict and fail, with the task's job, priority, CPU and memory request, run time, and the machine involved. 'Events' lists machine failures and returns, evictions and restarts, and the longest waits. 'Queue at window end' gives the state after the events at time 1200: the tasks still running (with their remaining run time) and the tasks still waiting, in queue order.

Tasks. A task belongs to a job and has a priority from 0 to 11 (higher is more important). Priorities 9 to 11 are the *production* tier and priorities 0 and 1 the *free* tier. Each task requests an amount of CPU and of memory, which it holds on one machine while it runs, and has a run time: once started it runs for exactly that many seconds and then finishes, unless it is stopped first. The run time is given on every row of the task log, starting with the task's submit row.

Scheduling policy. Every window starts with all machines empty and no task waiting. At every time at which something happens, in this order:

1. **Finishes.** Tasks whose run time is over finish and free their CPU and memory.

2. **Failures.** Machines that fail at this time go down. Each task running on such a machine stops (a *fail* event) and goes back to the queue.

3. **Returns.** Machines that return at this time come back, empty, with their capacity.

4. **Submissions.** Tasks submitted at this time join the queue.

5. **Scheduling round.** The waiting tasks are taken one at a time in *queue order*: higher priority first, then earlier submission time, then lower task id. Each task, in turn:

a. **First fit.** The task starts on the first machine, in machine id order, that is up and whose free CPU and free memory are both at least the task's requests (subject to the reservation rule in c).

b. **Preemption.** If it fits nowhere, no task holds a reservation yet in this round, and it is a production task, it may evict free-tier tasks from one machine. It looks at the up machines in id order. On each it considers the free-tier tasks running there, lowest priority first, then the most recently started first (then the higher task id first), and evicts them one by one in that order until it fits; it takes the first machine where this works, evicting only as many as needed there, and starts on it. If no machine works, nothing is evicted. An evicted task goes back to the queue at once (it may start again later in the same round); it loses its progress and will run its full run time again.

c. **Reservation and backfilling.** The first task in the round that can start neither by a. nor by b. gets a *reservation*: the earliest time at which it could start, computed from the run times of the tasks now running (assuming no new submissions, failures or returns), on the first machine, in id order, where that earliest time is reached. Every later task in the round may still start by first fit, but on the reserved machine only if it will finish no later than the reserved time, or if its requests fit in what the reserved machine will still have free at the reserved time after the reserved task's requests (and after other such starts on it in the same round). On the other machines it starts as usual. Later tasks do not preempt. A task that would not fit on any up machine even if that machine were empty gets no reservation and simply keeps waiting; the next task that cannot start may then get the reservation.

Tasks stopped by an eviction or a failure keep their original submission time (and so their place in queue order) and start again from the beginning. A task never moves while it runs.

Order of rows. Task log rows with the same time follow the steps above: finishes (by task id), failures (by machine, then task id), submissions (by task id), then the scheduling round's starts and evictions in the order the round makes them (each eviction just before the start that caused it).

Terms used in the Events section and in questions. A task's *wait* is the time from its submission to its first start. A task is *restarted* when it starts again after an eviction or a failure. The number of *evictions* counts evict events (one per evicted task). 'Running on a machine at time t' means running after all the events at time t.

The cluster and the workload. Each window has 8 machines, ids M01 to M08. Each is modelled on a real machine of the cell (one of them is a machine that failed during the window in the trace) and has 1/8 of that machine's recorded CPU and memory capacity. A failed machine returns at the time the trace records it coming back, if that is within 6 hours; otherwise it does not return. The tasks are a sample of the real tasks submitted in the cell during the window, chosen by this fixed rule: jobs submitted in the window are taken in a random order, first up to 6 production jobs, then free-tier jobs until the offered load reaches 54%, then other jobs until it reaches 90% (a job that would push it above 95% is skipped). A job contributes its first 4 tasks (16 for a free-tier job) that finished normally in the trace and fit on at least one machine of the window. Offered load = the larger, over CPU and memory, of the sum of request x run time inside the window over all tasks, divided by the cluster's total capacity x the window length.

After the window. For questions about what happens after time 1200: no new task is submitted and no machine fails after the window; a machine that is down returns at its listed return time (if any); the policy continues unchanged from the state given in 'Queue at window end'.

Units. Times and run times in seconds. CPU and memory in units of 1/100,000 of the largest machine in the original cell (the trace's normalised values, rounded to whole units).

Chapters in this document (itops_d02): Cell window 10, Cell window 14, Cell window 19, Cell window 21, Cell window 24.

Cell window 10

Times 0 to 1200. The cluster has 8 machines. 231 tasks of 111 jobs are submitted during the window.

Cell window 10: Machines

machine	CPU	memory	fails at	returns at
M01	6250	6244	-	-
M02	6250	9363	-	-
M03	6250	6244	-	-
M04	6250	6244	-	-
M05	6250	6244	-	-
M06	12500	12500	349	372
M07	6250	6244	-	-
M08	6250	3116	-	-

Cell window 10: Task log

time	task	job	priority	CPU	memory	run time	event	machine
3	T001	J01	6	625	187	21	submit	-
3	T001	J01	6	625	187	21	start	M01
14	T002	J02	1	1250	1553	183	submit	-
14	T002	J02	1	1250	1553	183	start	M01
24	T001	J01	6	625	187	21	finish	M01
34	T003	J03	1	1250	1553	77	submit	-
34	T003	J03	1	1250	1553	77	start	M01
38	T004	J04	1	1250	1553	329	submit	-
38	T004	J04	1	1250	1553	329	start	M01
40	T005	J05	0	1250	197	327	submit	-
40	T005	J05	0	1250	197	327	start	M01
42	T006	J06	0	1250	208	184	submit	-
42	T007	J06	0	1250	208	181	submit	-
42	T008	J06	0	1250	208	203	submit	-
42	T009	J06	0	1250	208	224	submit	-
42	T010	J06	0	1250	208	239	submit	-
42	T011	J06	0	1250	208	218	submit	-
42	T012	J06	0	1250	208	323	submit	-
42	T013	J06	0	1250	208	163	submit	-
42	T014	J06	0	1250	208	324	submit	-
42	T015	J06	0	1250	208	319	submit	-
42	T016	J06	0	1250	208	161	submit	-
42	T017	J06	0	1250	208	179	submit	-
42	T018	J06	0	1250	208	163	submit	-
42	T019	J06	0	1250	208	160	submit	-
42	T020	J06	0	1250	208	161	submit	-
42	T021	J06	0	1250	208	321	submit	-
42	T006	J06	0	1250	208	184	start	M01
42	T007	J06	0	1250	208	181	start	M02
42	T008	J06	0	1250	208	203	start	M02
42	T009	J06	0	1250	208	224	start	M02
42	T010	J06	0	1250	208	239	start	M02
42	T011	J06	0	1250	208	218	start	M02
42	T012	J06	0	1250	208	323	start	M03
42	T013	J06	0	1250	208	163	start	M03
42	T014	J06	0	1250	208	324	start	M03
42	T015	J06	0	1250	208	319	start	M03
42	T016	J06	0	1250	208	161	start	M03

time	task	job	priority	CPU	memory	run time	event	machine
42	T017	J06	0	1250	208	179	start	M04
42	T018	J06	0	1250	208	163	start	M04
42	T019	J06	0	1250	208	160	start	M04
42	T020	J06	0	1250	208	161	start	M04
42	T021	J06	0	1250	208	321	start	M04
56	T022	J07	1	1250	1553	257	submit	-
56	T022	J07	1	1250	1553	257	start	M05
59	T023	J08	4	3125	1125	42	submit	-
59	T023	J08	4	3125	1125	42	start	M05
72	T024	J09	6	625	3180	32	submit	-
72	T024	J09	6	625	3180	32	start	M05
97	T025	J10	6	625	3180	30	submit	-
97	T025	J10	6	625	3180	30	start	M06
101	T023	J08	4	3125	1125	42	finish	M05
104	T024	J09	6	625	3180	32	finish	M05
111	T003	J03	1	1250	1553	77	finish	M01
115	T026	J11	6	937	1244	244	submit	-
115	T026	J11	6	937	1244	244	start	M01
127	T025	J10	6	625	3180	30	finish	M06
186	T027	J12	4	62	398	642	submit	-
186	T027	J12	4	62	398	642	start	M01
188	T028	J13	4	1250	466	254	submit	-
188	T029	J13	4	1250	466	252	submit	-
188	T030	J13	4	1250	466	251	submit	-
188	T031	J13	4	1250	466	252	submit	-
188	T028	J13	4	1250	466	254	start	M05
188	T029	J13	4	1250	466	252	start	M05
188	T030	J13	4	1250	466	251	start	M05
188	T031	J13	4	1250	466	252	start	M05
197	T002	J02	1	1250	1553	183	finish	M01
198	T032	J14	6	625	3180	29	submit	-
198	T033	J15	8	3125	3180	159	submit	-
198	T033	J15	8	3125	3180	159	start	M06
198	T032	J14	6	625	3180	29	start	M06
202	T019	J06	0	1250	208	160	finish	M04
203	T016	J06	0	1250	208	161	finish	M03
203	T020	J06	0	1250	208	161	finish	M04
205	T013	J06	0	1250	208	163	finish	M03
205	T018	J06	0	1250	208	163	finish	M04
221	T017	J06	0	1250	208	179	finish	M04
223	T007	J06	0	1250	208	181	finish	M02
223	T034	J16	9	6250	3036	377	submit	-
223	T034	J16	9	6250	3036	377	start	M06
226	T006	J06	0	1250	208	184	finish	M01
227	T032	J14	6	625	3180	29	finish	M06
230	T035	J17	0	6250	466	45	submit	-
230	T035	J17	0	6250	466	45	start	M07
240	T036	J18	4	625	155	34	submit	-
240	T037	J19	4	625	155	34	submit	-
240	T038	J20	4	625	311	21	submit	-
240	T036	J18	4	625	155	34	start	M01
240	T037	J19	4	625	155	34	start	M01
240	T038	J20	4	625	311	21	start	M01
245	T008	J06	0	1250	208	203	finish	M02
252	T039	J21	6	625	3180	27	submit	-
252	T039	J21	6	625	3180	27	start	M02
260	T011	J06	0	1250	208	218	finish	M02

time	task	job	priority	CPU	memory	run time	event	machine
261	T038	J20	4	625	311	21	finish	M01
266	T009	J06	0	1250	208	224	finish	M02
274	T036	J18	4	625	155	34	finish	M01
274	T037	J19	4	625	155	34	finish	M01
275	T035	J17	0	6250	466	45	finish	M07
279	T039	J21	6	625	3180	27	finish	M02
281	T010	J06	0	1250	208	239	finish	M02
285	T040	J22	6	625	187	203	submit	-
285	T040	J22	6	625	187	203	start	M01
287	T041	J23	4	937	197	52	submit	-
287	T041	J23	4	937	197	52	start	M01
303	T042	J24	6	6250	622	150	submit	-
303	T043	J25	6	625	280	55	submit	-
303	T044	J26	8	4999	1590	4531	submit	-
303	T044	J26	8	4999	1590	4531	start	M02
303	T042	J24	6	6250	622	150	start	M07
303	T043	J25	6	625	280	55	start	M01
313	T022	J07	1	1250	1553	257	finish	M05
339	T041	J23	4	937	197	52	finish	M01
349	T033	J15	8	3125	3180	159	fail	M06
349	T034	J16	9	6250	3036	377	fail	M06
349	T034	J16	9	6250	3036	377	start	M08
349	T033	J15	8	3125	3180	159	start	M04
358	T043	J25	6	625	280	55	finish	M01
359	T026	J11	6	937	1244	244	finish	M01
361	T015	J06	0	1250	208	319	finish	M03
363	T021	J06	0	1250	208	321	finish	M04
365	T012	J06	0	1250	208	323	finish	M03
366	T014	J06	0	1250	208	324	finish	M03
367	T004	J04	1	1250	1553	329	finish	M01
367	T005	J05	0	1250	197	327	finish	M01
404	T045	J27	9	6250	1590	74	submit	-
404	T045	J27	9	6250	1590	74	start	M03
423	T046	J28	6	937	1244	87	submit	-
423	T046	J28	6	937	1244	87	start	M01
425	T047	J29	6	629	197	87	submit	-
425	T048	J29	6	629	197	83	submit	-
425	T049	J30	6	937	204	87	submit	-
425	T050	J30	6	937	204	76	submit	-
425	T051	J30	6	937	204	87	submit	-
425	T052	J30	6	937	204	85	submit	-
425	T047	J29	6	629	197	87	start	M01
425	T048	J29	6	629	197	83	start	M01
425	T049	J30	6	937	204	87	start	M01
425	T050	J30	6	937	204	76	start	M01
425	T051	J30	6	937	204	87	start	M01
425	T052	J30	6	937	204	85	start	M02
435	T053	J31	8	625	155	333	submit	-
435	T054	J32	8	625	155	461	submit	-
435	T053	J31	8	625	155	333	start	M04
435	T054	J32	8	625	155	461	start	M04
439	T030	J13	4	1250	466	251	finish	M05
440	T029	J13	4	1250	466	252	finish	M05
440	T031	J13	4	1250	466	252	finish	M05
442	T028	J13	4	1250	466	254	finish	M05
453	T042	J24	6	6250	622	150	finish	M07
459	T055	J33	1	1562	1553	165	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
459	T055	J33	1	1562	1553	165	start	M04
466	T056	J34	1	1562	1553	468	submit	-
466	T056	J34	1	1562	1553	468	start	M05
478	T045	J27	9	6250	1590	74	finish	M03
481	T057	J35	1	1562	1553	391	submit	-
481	T057	J35	1	1562	1553	391	start	M03
488	T040	J22	6	625	187	203	finish	M01
488	T058	J36	1	1562	1553	302	submit	-
488	T058	J36	1	1562	1553	302	start	M03
501	T050	J30	6	937	204	76	finish	M01
508	T033	J15	8	3125	3180	159	finish	M04
508	T048	J29	6	629	197	83	finish	M01
509	T059	J37	1	1250	1553	347	submit	-
509	T059	J37	1	1250	1553	347	start	M01
510	T046	J28	6	937	1244	87	finish	M01
510	T052	J30	6	937	204	85	finish	M02
511	T060	J38	0	1250	258	343	submit	-
511	T060	J38	0	1250	258	343	start	M01
512	T047	J29	6	629	197	87	finish	M01
512	T049	J30	6	937	204	87	finish	M01
512	T051	J30	6	937	204	87	finish	M01
513	T061	J39	0	1250	197	226	submit	-
513	T062	J39	0	1250	197	131	submit	-
513	T063	J39	0	1250	197	340	submit	-
513	T061	J39	0	1250	197	226	start	M01
513	T062	J39	0	1250	197	131	start	M01
513	T063	J39	0	1250	197	340	start	M02
516	T064	J40	4	625	225	58	submit	-
516	T065	J40	4	625	225	56	submit	-
516	T066	J40	4	625	225	54	submit	-
516	T067	J40	4	625	225	54	submit	-
516	T064	J40	4	625	225	58	start	M01
516	T065	J40	4	625	225	56	start	M03
516	T066	J40	4	625	225	54	start	M03
516	T067	J40	4	625	225	54	start	M03
525	T068	J41	1	1250	1553	251	submit	-
525	T068	J41	1	1250	1553	251	start	M03
526	T069	J42	0	1250	197	111	submit	-
526	T070	J42	0	1250	197	389	submit	-
526	T071	J42	0	1250	197	151	submit	-
526	T072	J42	0	1250	197	177	submit	-
526	T073	J42	0	1250	197	122	submit	-
526	T074	J42	0	1250	197	142	submit	-
526	T075	J42	0	1250	197	285	submit	-
526	T076	J42	0	1250	197	385	submit	-
526	T077	J42	0	1250	197	253	submit	-
526	T078	J42	0	1250	197	341	submit	-
526	T079	J42	0	1250	197	391	submit	-
526	T080	J42	0	1250	197	144	submit	-
526	T081	J42	0	1250	197	230	submit	-
526	T082	J42	0	1250	197	295	submit	-
526	T083	J42	0	1250	197	374	submit	-
526	T084	J42	0	1250	197	389	submit	-
526	T069	J42	0	1250	197	111	start	M04
526	T070	J42	0	1250	197	389	start	M04
526	T071	J42	0	1250	197	151	start	M05
526	T072	J42	0	1250	197	177	start	M05

time	task	job	priority	CPU	memory	run time	event	machine
526	T073	J42	0	1250	197	122	start	M05
526	T074	J42	0	1250	197	142	start	M06
526	T075	J42	0	1250	197	285	start	M06
526	T076	J42	0	1250	197	385	start	M06
526	T077	J42	0	1250	197	253	start	M06
526	T078	J42	0	1250	197	341	start	M06
526	T079	J42	0	1250	197	391	start	M06
526	T080	J42	0	1250	197	144	start	M06
526	T081	J42	0	1250	197	230	start	M06
526	T082	J42	0	1250	197	295	start	M06
526	T083	J42	0	1250	197	374	start	M06
526	T084	J42	0	1250	197	389	start	M07
529	T085	J43	9	6250	3180	214	submit	-
529	T073	J42	0	1250	197	122	evict	M05
529	T072	J42	0	1250	197	177	evict	M05
529	T071	J42	0	1250	197	151	evict	M05
529	T056	J34	1	1562	1553	468	evict	M05
529	T085	J43	9	6250	3180	214	start	M05
529	T056	J34	1	1562	1553	468	start	M07
529	T071	J42	0	1250	197	151	start	M07
529	T072	J42	0	1250	197	177	start	M07
532	T086	J44	0	1250	715	176	submit	-
534	T087	J45	1	1250	1553	78	submit	-
566	T088	J46	6	625	187	334	submit	-
566	T089	J47	6	6250	622	136	submit	-
566	T088	J46	6	625	187	334	start	M04
570	T066	J40	4	625	225	54	finish	M03
570	T067	J40	4	625	225	54	finish	M03
570	T073	J42	0	1250	197	122	start	M03
572	T065	J40	4	625	225	56	finish	M03
574	T064	J40	4	625	225	58	finish	M01
591	T090	J48	9	4688	6360	577	submit	-
591	T083	J42	0	1250	197	374	evict	M06
591	T082	J42	0	1250	197	295	evict	M06
591	T081	J42	0	1250	197	230	evict	M06
591	T080	J42	0	1250	197	144	evict	M06
591	T090	J48	9	4688	6360	577	start	M06
624	T055	J33	1	1562	1553	165	finish	M04
624	T087	J45	1	1250	1553	78	start	M04
637	T069	J42	0	1250	197	111	finish	M04
637	T080	J42	0	1250	197	144	start	M04
644	T062	J39	0	1250	197	131	finish	M01
644	T081	J42	0	1250	197	230	start	M01
668	T074	J42	0	1250	197	142	finish	M06
668	T082	J42	0	1250	197	295	start	M06
672	T091	J49	6	625	187	30	submit	-
672	T092	J50	8	937	466	65	submit	-
672	T092	J50	8	937	466	65	start	M01
672	T091	J49	6	625	187	30	start	M03
674	T093	J51	4	625	197	64	submit	-
674	T093	J51	4	625	197	64	start	M04
680	T071	J42	0	1250	197	151	finish	M07
680	T083	J42	0	1250	197	374	start	M07
692	T073	J42	0	1250	197	122	finish	M03
692	T086	J44	0	1250	715	176	start	M03
702	T087	J45	1	1250	1553	78	finish	M04
702	T091	J49	6	625	187	30	finish	M03

time	task	job	priority	CPU	memory	run time	event	machine
706	T072	J42	0	1250	197	177	finish	M07
725	T094	J52	4	62	398	55	submit	-
725	T094	J52	4	62	398	55	start	M01
726	T034	J16	9	6250	3036	377	finish	M08
726	T089	J47	6	6250	622	136	start	M08
737	T092	J50	8	937	466	65	finish	M01
738	T093	J51	4	625	197	64	finish	M04
739	T061	J39	0	1250	197	226	finish	M01
740	T095	J53	1	1250	1553	50	submit	-
740	T095	J53	1	1250	1553	50	start	M01
742	T096	J54	0	1250	197	48	submit	-
742	T096	J54	0	1250	197	48	start	M04
743	T085	J43	9	6250	3180	214	finish	M05
768	T053	J31	8	625	155	333	finish	M04
771	T097	J55	4	625	217	67	submit	-
771	T098	J55	4	625	217	67	submit	-
771	T099	J55	4	625	217	72	submit	-
771	T100	J55	4	625	217	67	submit	-
771	T097	J55	4	625	217	67	start	M01
771	T098	J55	4	625	217	67	start	M03
771	T099	J55	4	625	217	72	start	M04
771	T100	J55	4	625	217	67	start	M04
776	T068	J41	1	1250	1553	251	finish	M03
776	T101	J56	1	1562	1553	365	submit	-
776	T101	J56	1	1562	1553	365	start	M05
779	T077	J42	0	1250	197	253	finish	M06
780	T094	J52	4	62	398	55	finish	M01
781	T080	J42	0	1250	197	144	finish	M04
782	T102	J57	8	1250	777	70	submit	-
782	T103	J58	8	625	155	88	submit	-
782	T102	J57	8	1250	777	70	start	M03
782	T103	J58	8	625	155	88	start	M04
790	T058	J36	1	1562	1553	302	finish	M03
790	T095	J53	1	1250	1553	50	finish	M01
790	T096	J54	0	1250	197	48	finish	M04
809	T104	J59	1	1639	1553	313	submit	-
809	T104	J59	1	1639	1553	313	start	M01
811	T075	J42	0	1250	197	285	finish	M06
811	T105	J60	0	736	317	311	submit	-
811	T106	J61	4	1250	466	344	submit	-
811	T106	J61	4	1250	466	344	start	M03
811	T105	J60	0	736	317	311	start	M04
813	T107	J62	4	1250	466	235	submit	-
813	T108	J62	4	1250	466	242	submit	-
813	T109	J62	4	1250	466	246	submit	-
813	T110	J62	4	1250	466	250	submit	-
813	T111	J63	4	1250	466	266	submit	-
813	T112	J63	4	1250	466	284	submit	-
813	T113	J63	4	1250	466	224	submit	-
813	T114	J63	4	1250	466	222	submit	-
813	T115	J64	0	1250	835	148	submit	-
813	T116	J64	0	1250	835	200	submit	-
813	T117	J64	0	1250	835	218	submit	-
813	T118	J64	0	1250	835	167	submit	-
813	T119	J64	0	1250	835	143	submit	-
813	T120	J64	0	1250	835	175	submit	-
813	T121	J64	0	1250	835	196	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
813	T122	J64	0	1250	835	136	submit	-
813	T123	J64	0	1250	835	199	submit	-
813	T124	J64	0	1250	835	125	submit	-
813	T125	J64	0	1250	835	147	submit	-
813	T126	J64	0	1250	835	217	submit	-
813	T127	J64	0	1250	835	308	submit	-
813	T128	J64	0	1250	835	234	submit	-
813	T129	J64	0	1250	835	144	submit	-
813	T130	J64	0	1250	835	306	submit	-
813	T107	J62	4	1250	466	235	start	M05
813	T108	J62	4	1250	466	242	start	M05
813	T109	J62	4	1250	466	246	start	M05
813	T110	J62	4	1250	466	250	start	M06
813	T111	J63	4	1250	466	266	start	M06
813	T112	J63	4	1250	466	284	start	M07
823	T131	J65	1	1562	1553	102	submit	-
828	T027	J12	4	62	398	642	finish	M01
832	T132	J66	0	6250	466	44	submit	-
838	T097	J55	4	625	217	67	finish	M01
838	T098	J55	4	625	217	67	finish	M03
838	T100	J55	4	625	217	67	finish	M04
838	T113	J63	4	1250	466	224	start	M04
840	T133	J67	1	3125	2386	245	submit	-
842	T134	J68	4	3125	777	371	submit	-
842	T135	J68	4	3125	777	358	submit	-
842	T136	J68	4	3125	777	363	submit	-
842	T137	J68	4	3125	777	359	submit	-
842	T138	J69	4	625	155	40	submit	-
842	T139	J70	4	625	311	39	submit	-
842	T140	J71	8	1874	199	114	submit	-
842	T138	J69	4	625	155	40	start	M01
842	T139	J70	4	625	311	39	start	M05
843	T099	J55	4	625	217	72	finish	M04
852	T102	J57	8	1250	777	70	finish	M03
852	T140	J71	8	1874	199	114	start	M03
853	T063	J39	0	1250	197	340	finish	M02
853	T114	J63	4	1250	466	222	start	M02
854	T060	J38	0	1250	258	343	finish	M01
854	T115	J64	0	1250	835	148	start	M01
856	T059	J37	1	1250	1553	347	finish	M01
856	T116	J64	0	1250	835	200	start	M01
857	T141	J72	1	1250	1553	53	submit	-
859	T142	J73	0	1250	197	49	submit	-
860	T143	J74	4	1250	398	33	submit	-
860	T144	J75	8	6250	1590	1278	submit	-
860	T145	J76	4	3125	311	320	submit	-
862	T089	J47	6	6250	622	136	finish	M08
862	T144	J75	8	6250	1590	1278	start	M08
864	T146	J77	1	1250	1553	46	submit	-
867	T078	J42	0	1250	197	341	finish	M06
867	T143	J74	4	1250	398	33	start	M06
868	T086	J44	0	1250	715	176	finish	M03
870	T103	J58	8	625	155	88	finish	M04
870	T131	J65	1	1562	1553	102	start	M04
872	T057	J35	1	1562	1553	391	finish	M03
872	T134	J68	4	3125	777	371	start	M03
873	T147	J78	1	1250	466	188	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
874	T081	J42	0	1250	197	230	finish	M01
874	T141	J72	1	1250	1553	53	start	M01
876	T148	J79	0	625	219	186	submit	-
876	T149	J79	0	625	219	170	submit	-
876	T150	J79	0	625	219	170	submit	-
876	T151	J79	0	625	219	186	submit	-
876	T152	J79	0	625	219	174	submit	-
876	T153	J79	0	625	219	170	submit	-
876	T154	J79	0	625	219	176	submit	-
876	T155	J79	0	625	219	175	submit	-
876	T156	J79	0	625	219	186	submit	-
876	T157	J79	0	625	219	182	submit	-
876	T158	J79	0	625	219	182	submit	-
876	T159	J79	0	625	219	183	submit	-
876	T160	J79	0	625	219	183	submit	-
876	T161	J79	0	625	219	170	submit	-
876	T162	J79	0	625	219	170	submit	-
876	T163	J79	0	625	219	174	submit	-
876	T148	J79	0	625	219	186	start	M07
881	T139	J70	4	625	311	39	finish	M05
881	T149	J79	0	625	219	170	start	M05
882	T138	J69	4	625	155	40	finish	M01
882	T150	J79	0	625	219	170	start	M01
896	T054	J32	8	625	155	461	finish	M04
896	T151	J79	0	625	219	186	start	M04
900	T088	J46	6	625	187	334	finish	M04
900	T143	J74	4	1250	398	33	finish	M06
900	T152	J79	0	625	219	174	start	M04
900	T153	J79	0	625	219	170	start	M06
901	T164	J80	1	1562	1553	116	submit	-
902	T165	J81	4	3125	1125	48	submit	-
911	T076	J42	0	1250	197	385	finish	M06
915	T070	J42	0	1250	197	389	finish	M04
915	T084	J42	0	1250	197	389	finish	M07
915	T146	J77	1	1250	1553	46	start	M04
915	T147	J78	1	1250	466	188	start	M07
917	T079	J42	0	1250	197	391	finish	M06
917	T135	J68	4	3125	777	358	start	M06
921	T166	J82	6	625	280	330	submit	-
921	T167	J83	8	1250	777	61	submit	-
927	T141	J72	1	1250	1553	53	finish	M01
927	T167	J83	8	1250	777	61	start	M01
939	T168	J84	6	937	1244	84	submit	-
941	T169	J85	6	937	202	83	submit	-
961	T146	J77	1	1250	1553	46	finish	M04
961	T166	J82	6	625	280	330	start	M04
961	T154	J79	0	625	219	176	start	M04
963	T082	J42	0	1250	197	295	finish	M06
963	T168	J84	6	937	1244	84	start	M06
963	T155	J79	0	625	219	175	start	M06
966	T140	J71	8	1874	199	114	finish	M03
966	T169	J85	6	937	202	83	start	M03
966	T156	J79	0	625	219	186	start	M03
972	T131	J65	1	1562	1553	102	finish	M04
972	T164	J80	1	1562	1553	116	start	M04
977	T170	J86	6	937	280	126	submit	-
977	T171	J86	6	937	280	59	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
977	T172	J86	6	937	280	76	submit	-
977	T173	J86	6	937	280	90	submit	-
981	T174	J87	1	1562	1553	325	submit	-
988	T167	J83	8	1250	777	61	finish	M01
988	T170	J86	6	937	280	126	start	M01
997	T056	J34	1	1562	1553	468	finish	M07
997	T171	J86	6	937	280	59	start	M07
997	T172	J86	6	937	280	76	start	M07
1002	T115	J64	0	1250	835	148	finish	M01
1002	T175	J88	1	1562	1553	301	submit	-
1002	T173	J86	6	937	280	90	start	M01
1002	T157	J79	0	625	219	182	start	M01
1033	T176	J89	1	1250	1553	256	submit	-
1033	T177	J90	1	1250	1553	52	submit	-
1035	T178	J91	0	905	227	254	submit	-
1035	T179	J92	0	1250	197	50	submit	-
1037	T180	J93	1	1250	1553	254	submit	-
1039	T181	J94	0	1250	283	252	submit	-
1041	T182	J95	0	1250	217	147	submit	-
1041	T183	J95	0	1250	217	223	submit	-
1041	T184	J95	0	1250	217	188	submit	-
1041	T185	J95	0	1250	217	168	submit	-
1041	T186	J95	0	1250	217	205	submit	-
1041	T187	J95	0	1250	217	246	submit	-
1041	T188	J95	0	1250	217	211	submit	-
1041	T189	J95	0	1250	217	183	submit	-
1041	T190	J95	0	1250	217	178	submit	-
1041	T191	J95	0	1250	217	164	submit	-
1041	T192	J95	0	1250	217	201	submit	-
1041	T193	J95	0	1250	217	247	submit	-
1041	T194	J95	0	1250	217	231	submit	-
1041	T195	J95	0	1250	217	248	submit	-
1042	T196	J96	6	625	3180	32	submit	-
1047	T168	J84	6	937	1244	84	finish	M06
1047	T196	J96	6	625	3180	32	start	M06
1048	T107	J62	4	1250	466	235	finish	M05
1049	T169	J85	6	937	202	83	finish	M03
1049	T176	J89	1	1250	1553	256	start	M03
1051	T149	J79	0	625	219	170	finish	M05
1052	T150	J79	0	625	219	170	finish	M01
1052	T158	J79	0	625	219	182	start	M01
1054	T083	J42	0	1250	197	374	finish	M07
1054	T177	J90	1	1250	1553	52	start	M07
1055	T108	J62	4	1250	466	242	finish	M05
1055	T136	J68	4	3125	777	363	start	M05
1056	T116	J64	0	1250	835	200	finish	M01
1056	T171	J86	6	937	280	59	finish	M07
1056	T180	J93	1	1250	1553	254	start	M01
1056	T159	J79	0	625	219	183	start	M07
1059	T109	J62	4	1250	466	246	finish	M05
1059	T174	J87	1	1562	1553	325	start	M05
1062	T113	J63	4	1250	466	224	finish	M04
1062	T148	J79	0	625	219	186	finish	M07
1062	T117	J64	0	1250	835	218	start	M04
1062	T160	J79	0	625	219	183	start	M07
1063	T110	J62	4	1250	466	250	finish	M06
1063	T161	J79	0	625	219	170	start	M06

time	task	job	priority	CPU	memory	run time	event	machine
1070	T153	J79	0	625	219	170	finish	M06
1070	T197	J97	0	6250	1244	77	submit	-
1073	T172	J86	6	937	280	76	finish	M07
1073	T118	J64	0	1250	835	167	start	M07
1074	T152	J79	0	625	219	174	finish	M04
1074	T162	J79	0	625	219	170	start	M04
1075	T114	J63	4	1250	466	222	finish	M02
1075	T119	J64	0	1250	835	143	start	M02
1079	T111	J63	4	1250	466	266	finish	M06
1079	T196	J96	6	625	3180	32	finish	M06
1079	T137	J68	4	3125	777	359	start	M06
1082	T151	J79	0	625	219	186	finish	M04
1082	T198	J98	1	1562	1553	227	submit	-
1082	T163	J79	0	625	219	174	start	M04
1085	T199	J99	4	625	155	24	submit	-
1088	T164	J80	1	1562	1553	116	finish	M04
1088	T199	J99	4	625	155	24	start	M04
1088	T178	J91	0	905	227	254	start	M04
1092	T173	J86	6	937	280	90	finish	M01
1097	T112	J63	4	1250	466	284	finish	M07
1103	T147	J78	1	1250	466	188	finish	M07
1103	T200	J100	8	625	1553	1276	submit	-
1103	T201	J101	8	625	311	374	submit	-
1103	T200	J100	8	625	1553	1276	start	M01
1103	T201	J101	8	625	311	374	start	M07
1106	T177	J90	1	1250	1553	52	finish	M07
1106	T145	J76	4	3125	311	320	start	M07
1112	T199	J99	4	625	155	24	finish	M04
1114	T170	J86	6	937	280	126	finish	M01
1122	T104	J59	1	1639	1553	313	finish	M01
1122	T105	J60	0	736	317	311	finish	M04
1122	T165	J81	4	3125	1125	48	start	M01
1122	T175	J88	1	1562	1553	301	start	M04
1129	T202	J102	1	1250	1553	169	submit	-
1129	T203	J103	0	1250	16	27	submit	-
1130	T204	J104	0	1250	204	167	submit	-
1137	T154	J79	0	625	219	176	finish	M04
1138	T155	J79	0	625	219	175	finish	M06
1141	T101	J56	1	1562	1553	365	finish	M05
1141	T198	J98	1	1562	1553	227	start	M05
1147	T205	J105	4	625	155	35	submit	-
1147	T205	J105	4	625	155	35	start	M04
1149	T206	J106	4	1250	466	264	submit	-
1149	T207	J106	4	1250	466	290	submit	-
1149	T208	J106	4	1250	466	267	submit	-
1149	T209	J106	4	1250	466	266	submit	-
1152	T156	J79	0	625	219	186	finish	M03
1155	T106	J61	4	1250	466	344	finish	M03
1155	T206	J106	4	1250	466	264	start	M03
1160	T210	J107	1	1562	1553	633	submit	-
1165	T211	J108	4	6250	622	463	submit	-
1165	T212	J108	4	6250	622	423	submit	-
1165	T213	J108	4	6250	622	430	submit	-
1165	T214	J108	4	6250	622	364	submit	-
1165	T215	J109	4	1477	197	297	submit	-
1165	T216	J109	4	1477	197	282	submit	-
1165	T217	J109	4	1477	197	278	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
1165	T218	J109	4	1477	197	278	submit	-
1168	T090	J48	9	4688	6360	577	finish	M06
1168	T207	J106	4	1250	466	290	start	M06
1168	T208	J106	4	1250	466	267	start	M06
1168	T209	J106	4	1250	466	266	start	M06
1168	T215	J109	4	1477	197	297	start	M06
1170	T165	J81	4	3125	1125	48	finish	M01
1170	T219	J110	9	6250	1590	86	submit	-
1170	T216	J109	4	1477	197	282	start	M01
1170	T217	J109	4	1477	197	278	start	M01
1182	T205	J105	4	625	155	35	finish	M04
1183	T220	J111	0	1250	610	1134	submit	-
1183	T221	J111	0	1250	610	1124	submit	-
1183	T222	J111	0	1250	610	1125	submit	-
1183	T223	J111	0	1250	610	1122	submit	-
1183	T224	J111	0	1250	610	1122	submit	-
1183	T225	J111	0	1250	610	1121	submit	-
1183	T226	J111	0	1250	610	1122	submit	-
1183	T227	J111	0	1250	610	1123	submit	-
1183	T228	J111	0	1250	610	1122	submit	-
1183	T229	J111	0	1250	610	1121	submit	-
1183	T230	J111	0	1250	610	1123	submit	-
1183	T231	J111	0	1250	610	81	submit	-
1184	T157	J79	0	625	219	182	finish	M01

Cell window 10: Events

Time 349: machine M06 failed; tasks T033, T034 stopped and went back to the queue.

Time 349: task T033 restarted on machine M04.

Time 349: task T034 restarted on machine M08.

Time 372: machine M06 returned.

Time 529: task T085 (priority 9) evicted task T056 (priority 1) from machine M05.

Time 529: task T085 (priority 9) evicted task T071 (priority 0) from machine M05.

Time 529: task T085 (priority 9) evicted task T072 (priority 0) from machine M05.

Time 529: task T085 (priority 9) evicted task T073 (priority 0) from machine M05.

Time 529: task T056 restarted on machine M07.

Time 529: task T071 restarted on machine M07.

Time 529: task T072 restarted on machine M07.

Time 570: task T073 restarted on machine M03.

Time 591: task T090 (priority 9) evicted task T080 (priority 0) from machine M06.

Time 591: task T090 (priority 9) evicted task T081 (priority 0) from machine M06.

Time 591: task T090 (priority 9) evicted task T082 (priority 0) from machine M06.

Time 591: task T090 (priority 9) evicted task T083 (priority 0) from machine M06.

Time 637: task T080 restarted on machine M04.

Time 644: task T081 restarted on machine M01.

Time 668: task T082 restarted on machine M06.

Time 680: task T083 restarted on machine M07.

Longest waits (submission to first start, tasks started by the window end): task T119 waited 262 s (submitted at 813, first started at 1075); task T118 waited 260 s (submitted at 813, first started at 1073); task T117 waited 249 s (submitted at 813, first started at 1062); task T145 waited 246 s (submitted at 860, first started at 1106); task T137 waited 237 s (submitted at 842, first started at 1079).

Cell window 10: Queue at window end

State after the events at time 1200.

Tasks running at time 1200 (remaining run time = seconds still to run after time 1200):

task	job	priority	CPU	memory	machine	started	remaining
T158	J79	0	625	219	M01	1052	34
T180	J93	1	1250	1553	M01	1056	110
T200	J100	8	625	1553	M01	1103	1179
T216	J109	4	1477	197	M01	1170	252
T217	J109	4	1477	197	M01	1170	248
T044	J26	8	4999	1590	M02	303	3634
T119	J64	0	1250	835	M02	1075	18
T134	J68	4	3125	777	M03	872	43
T176	J89	1	1250	1553	M03	1049	105
T206	J106	4	1250	466	M03	1155	219
T117	J64	0	1250	835	M04	1062	80
T162	J79	0	625	219	M04	1074	44
T163	J79	0	625	219	M04	1082	56
T166	J82	6	625	280	M04	961	91
T175	J88	1	1562	1553	M04	1122	223
T178	J91	0	905	227	M04	1088	142
T136	J68	4	3125	777	M05	1055	218
T174	J87	1	1562	1553	M05	1059	184
T198	J98	1	1562	1553	M05	1141	168
T135	J68	4	3125	777	M06	917	75
T137	J68	4	3125	777	M06	1079	238
T161	J79	0	625	219	M06	1063	33
T207	J106	4	1250	466	M06	1168	258
T208	J106	4	1250	466	M06	1168	235
T209	J106	4	1250	466	M06	1168	234
T215	J109	4	1477	197	M06	1168	265
T118	J64	0	1250	835	M07	1073	40
T145	J76	4	3125	311	M07	1106	226
T159	J79	0	625	219	M07	1056	39
T160	J79	0	625	219	M07	1062	45
T201	J101	8	625	311	M07	1103	277
T144	J75	8	6250	1590	M08	862	940

Tasks waiting at time 1200, in queue order (each will run its full run time once started):

task	job	priority	CPU	memory	submitted	run time
T219	J110	9	6250	1590	1170	86
T211	J108	4	6250	622	1165	463
T212	J108	4	6250	622	1165	423
T213	J108	4	6250	622	1165	430
T214	J108	4	6250	622	1165	364
T218	J109	4	1477	197	1165	278
T133	J67	1	3125	2386	840	245
T202	J102	1	1250	1553	1129	169
T210	J107	1	1562	1553	1160	633
T120	J64	0	1250	835	813	175

task	job	priority	CPU	memory	submitted	run time
T121	J64	0	1250	835	813	196
T122	J64	0	1250	835	813	136
T123	J64	0	1250	835	813	199
T124	J64	0	1250	835	813	125
T125	J64	0	1250	835	813	147
T126	J64	0	1250	835	813	217
T127	J64	0	1250	835	813	308
T128	J64	0	1250	835	813	234
T129	J64	0	1250	835	813	144
T130	J64	0	1250	835	813	306
T132	J66	0	6250	466	832	44
T142	J73	0	1250	197	859	49
T179	J92	0	1250	197	1035	50
T181	J94	0	1250	283	1039	252
T182	J95	0	1250	217	1041	147
T183	J95	0	1250	217	1041	223
T184	J95	0	1250	217	1041	188
T185	J95	0	1250	217	1041	168
T186	J95	0	1250	217	1041	205
T187	J95	0	1250	217	1041	246
T188	J95	0	1250	217	1041	211
T189	J95	0	1250	217	1041	183
T190	J95	0	1250	217	1041	178
T191	J95	0	1250	217	1041	164
T192	J95	0	1250	217	1041	201
T193	J95	0	1250	217	1041	247
T194	J95	0	1250	217	1041	231
T195	J95	0	1250	217	1041	248
T197	J97	0	6250	1244	1070	77
T203	J103	0	1250	16	1129	27
T204	J104	0	1250	204	1130	167
T220	J111	0	1250	610	1183	1134
T221	J111	0	1250	610	1183	1124
T222	J111	0	1250	610	1183	1125
T223	J111	0	1250	610	1183	1122
T224	J111	0	1250	610	1183	1122
T225	J111	0	1250	610	1183	1121
T226	J111	0	1250	610	1183	1122
T227	J111	0	1250	610	1183	1123
T228	J111	0	1250	610	1183	1122
T229	J111	0	1250	610	1183	1121
T230	J111	0	1250	610	1183	1123
T231	J111	0	1250	610	1183	81

All machines are up at time 1200.

Cell window 14

Times 0 to 1200. The cluster has 8 machines. 183 tasks of 75 jobs are submitted during the window.

Cell window 14: Machines

machine	CPU	memory	fails at	returns at
M01	6250	3116	-	-
M02	6250	3116	-	-
M03	6250	3116	-	-
M04	6250	6244	-	-
M05	6250	6244	585	7370
M06	6250	6244	-	-
M07	6250	3116	-	-
M08	6250	3116	-	-

Cell window 14: Task log

time	task	job	priority	CPU	memory	run time	event	machine
0	T001	J01	0	736	317	311	submit	-
0	T001	J01	0	736	317	311	start	M01
2	T002	J02	4	1250	466	235	submit	-
2	T003	J02	4	1250	466	242	submit	-
2	T004	J02	4	1250	466	246	submit	-
2	T005	J02	4	1250	466	250	submit	-
2	T006	J03	4	1250	466	266	submit	-
2	T007	J03	4	1250	466	284	submit	-
2	T008	J03	4	1250	466	224	submit	-
2	T009	J03	4	1250	466	222	submit	-
2	T010	J04	0	1250	835	148	submit	-
2	T011	J04	0	1250	835	200	submit	-
2	T012	J04	0	1250	835	218	submit	-
2	T013	J04	0	1250	835	167	submit	-
2	T014	J04	0	1250	835	143	submit	-
2	T015	J04	0	1250	835	175	submit	-
2	T016	J04	0	1250	835	196	submit	-
2	T017	J04	0	1250	835	136	submit	-
2	T018	J04	0	1250	835	199	submit	-
2	T019	J04	0	1250	835	125	submit	-
2	T020	J04	0	1250	835	147	submit	-
2	T021	J04	0	1250	835	217	submit	-
2	T022	J04	0	1250	835	308	submit	-
2	T023	J04	0	1250	835	234	submit	-
2	T024	J04	0	1250	835	144	submit	-
2	T025	J04	0	1250	835	306	submit	-
2	T002	J02	4	1250	466	235	start	M01
2	T003	J02	4	1250	466	242	start	M01
2	T004	J02	4	1250	466	246	start	M01
2	T005	J02	4	1250	466	250	start	M01
2	T006	J03	4	1250	466	266	start	M02
2	T007	J03	4	1250	466	284	start	M02
2	T008	J03	4	1250	466	224	start	M02
2	T009	J03	4	1250	466	222	start	M02
2	T010	J04	0	1250	835	148	start	M02
2	T011	J04	0	1250	835	200	start	M03
2	T012	J04	0	1250	835	218	start	M03
2	T013	J04	0	1250	835	167	start	M03

time	task	job	priority	CPU	memory	run time	event	machine
2	T014	J04	0	1250	835	143	start	M04
2	T015	J04	0	1250	835	175	start	M04
2	T016	J04	0	1250	835	196	start	M04
2	T017	J04	0	1250	835	136	start	M04
2	T018	J04	0	1250	835	199	start	M04
2	T019	J04	0	1250	835	125	start	M05
2	T020	J04	0	1250	835	147	start	M05
2	T021	J04	0	1250	835	217	start	M05
2	T022	J04	0	1250	835	308	start	M05
2	T023	J04	0	1250	835	234	start	M05
2	T024	J04	0	1250	835	144	start	M06
2	T025	J04	0	1250	835	306	start	M06
5	T026	J05	1	1562	1553	105	submit	-
5	T026	J05	1	1562	1553	105	start	M06
25	T027	J06	1	1250	1590	432	submit	-
25	T027	J06	1	1250	1590	432	start	M06
31	T028	J07	4	625	311	240	submit	-
31	T029	J08	4	625	311	39	submit	-
31	T028	J07	4	625	311	240	start	M03
31	T029	J08	4	625	311	39	start	M06
46	T030	J09	1	1250	1553	53	submit	-
46	T030	J09	1	1250	1553	53	start	M07
48	T031	J10	0	1250	197	49	submit	-
48	T031	J10	0	1250	197	49	start	M03
49	T032	J11	4	1250	398	33	submit	-
49	T033	J12	8	6250	1590	1278	submit	-
49	T033	J12	8	6250	1590	1278	start	M08
49	T032	J11	4	1250	398	33	start	M07
53	T034	J13	1	1250	1553	46	submit	-
62	T035	J14	1	1250	466	188	submit	-
65	T036	J15	0	625	219	186	submit	-
65	T037	J15	0	625	219	170	submit	-
65	T038	J15	0	625	219	170	submit	-
65	T039	J15	0	625	219	186	submit	-
65	T040	J15	0	625	219	174	submit	-
65	T041	J15	0	625	219	170	submit	-
65	T042	J15	0	625	219	176	submit	-
65	T043	J15	0	625	219	175	submit	-
65	T044	J15	0	625	219	186	submit	-
65	T045	J15	0	625	219	182	submit	-
65	T046	J15	0	625	219	182	submit	-
65	T047	J15	0	625	219	183	submit	-
65	T048	J15	0	625	219	183	submit	-
65	T049	J15	0	625	219	170	submit	-
65	T050	J15	0	625	219	170	submit	-
65	T051	J15	0	625	219	174	submit	-
70	T029	J08	4	625	311	39	finish	M06
70	T036	J15	0	625	219	186	start	M06
74	T052	J16	6	625	280	151	submit	-
74	T052	J16	6	625	280	151	start	M07
74	T035	J14	1	1250	466	188	start	M07
74	T037	J15	0	625	219	170	start	M07
82	T032	J11	4	1250	398	33	finish	M07
82	T038	J15	0	625	219	170	start	M07
82	T039	J15	0	625	219	186	start	M07
90	T053	J17	1	1562	1553	116	submit	-
97	T031	J10	0	1250	197	49	finish	M03

time	task	job	priority	CPU	memory	run time	event	machine
97	T040	J15	0	625	219	174	start	M03
99	T030	J09	1	1250	1553	53	finish	M07
99	T034	J13	1	1250	1553	46	start	M07
110	T026	J05	1	1562	1553	105	finish	M06
110	T054	J18	4	937	1035	113	submit	-
110	T055	J18	4	937	1035	82	submit	-
110	T056	J18	4	937	1035	81	submit	-
110	T057	J18	4	937	1035	98	submit	-
110	T058	J19	8	1250	777	61	submit	-
110	T058	J19	8	1250	777	61	start	M06
110	T041	J15	0	625	219	170	start	M06
127	T019	J04	0	1250	835	125	finish	M05
127	T054	J18	4	937	1035	113	start	M05
128	T059	J20	6	937	1244	84	submit	-
130	T060	J21	6	937	202	83	submit	-
138	T017	J04	0	1250	835	136	finish	M04
138	T059	J20	6	937	1244	84	start	M04
145	T014	J04	0	1250	835	143	finish	M04
145	T034	J13	1	1250	1553	46	finish	M07
145	T060	J21	6	937	202	83	start	M04
145	T055	J18	4	937	1035	82	start	M07
145	T042	J15	0	625	219	176	start	M04
145	T043	J15	0	625	219	175	start	M07
145	T044	J15	0	625	219	186	start	M07
146	T024	J04	0	1250	835	144	finish	M06
146	T056	J18	4	937	1035	81	start	M06
148	T061	J22	6	937	1244	277	submit	-
149	T020	J04	0	1250	835	147	finish	M05
149	T061	J22	6	937	1244	277	start	M05
149	T045	J15	0	625	219	182	start	M05
150	T010	J04	0	1250	835	148	finish	M02
150	T057	J18	4	937	1035	98	start	M02
155	T062	J23	1	1250	1553	37	submit	-
169	T013	J04	0	1250	835	167	finish	M03
169	T046	J15	0	625	219	182	start	M03
169	T047	J15	0	625	219	183	start	M03
169	T048	J15	0	625	219	183	start	M03
169	T049	J15	0	625	219	170	start	M03
170	T063	J24	1	1562	1553	325	submit	-
171	T058	J19	8	1250	777	61	finish	M06
171	T053	J17	1	1562	1553	116	start	M06
177	T015	J04	0	1250	835	175	finish	M04
177	T062	J23	1	1250	1553	37	start	M04
184	T064	J25	1	1562	1553	97	submit	-
198	T016	J04	0	1250	835	196	finish	M04
198	T050	J15	0	625	219	170	start	M04
201	T018	J04	0	1250	835	199	finish	M04
201	T063	J24	1	1562	1553	325	start	M04
202	T011	J04	0	1250	835	200	finish	M03
202	T051	J15	0	625	219	174	start	M03
214	T062	J23	1	1250	1553	37	finish	M04
214	T064	J25	1	1562	1553	97	start	M04
219	T021	J04	0	1250	835	217	finish	M05
220	T012	J04	0	1250	835	218	finish	M03
222	T059	J20	6	937	1244	84	finish	M04
222	T065	J26	1	1250	1553	256	submit	-
222	T066	J27	1	1250	1553	52	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
222	T065	J26	1	1250	1553	256	start	M05
224	T009	J03	4	1250	466	222	finish	M02
224	T067	J28	0	905	227	254	submit	-
224	T068	J29	0	1250	197	50	submit	-
224	T067	J28	0	905	227	254	start	M02
224	T068	J29	0	1250	197	50	start	M03
225	T052	J16	6	625	280	151	finish	M07
226	T008	J03	4	1250	466	224	finish	M02
227	T055	J18	4	937	1035	82	finish	M07
227	T056	J18	4	937	1035	81	finish	M06
227	T066	J27	1	1250	1553	52	start	M07
228	T060	J21	6	937	202	83	finish	M04
228	T069	J30	0	1250	283	252	submit	-
228	T069	J30	0	1250	283	252	start	M02
230	T070	J31	0	1250	217	147	submit	-
230	T071	J31	0	1250	217	223	submit	-
230	T072	J31	0	1250	217	188	submit	-
230	T073	J31	0	1250	217	168	submit	-
230	T074	J31	0	1250	217	205	submit	-
230	T075	J31	0	1250	217	246	submit	-
230	T076	J31	0	1250	217	211	submit	-
230	T077	J31	0	1250	217	183	submit	-
230	T078	J31	0	1250	217	178	submit	-
230	T079	J31	0	1250	217	164	submit	-
230	T080	J31	0	1250	217	201	submit	-
230	T081	J31	0	1250	217	247	submit	-
230	T082	J31	0	1250	217	231	submit	-
230	T083	J31	0	1250	217	248	submit	-
230	T070	J31	0	1250	217	147	start	M04
236	T023	J04	0	1250	835	234	finish	M05
236	T071	J31	0	1250	217	223	start	M05
237	T002	J02	4	1250	466	235	finish	M01
237	T072	J31	0	1250	217	188	start	M01
240	T054	J18	4	937	1035	113	finish	M05
244	T003	J02	4	1250	466	242	finish	M01
244	T037	J15	0	625	219	170	finish	M07
244	T073	J31	0	1250	217	168	start	M01
244	T074	J31	0	1250	217	205	start	M07
248	T004	J02	4	1250	466	246	finish	M01
248	T057	J18	4	937	1035	98	finish	M02
248	T075	J31	0	1250	217	246	start	M01
248	T076	J31	0	1250	217	211	start	M02
252	T005	J02	4	1250	466	250	finish	M01
252	T038	J15	0	625	219	170	finish	M07
252	T077	J31	0	1250	217	183	start	M01
256	T036	J15	0	625	219	186	finish	M06
256	T078	J31	0	1250	217	178	start	M06
262	T035	J14	1	1250	466	188	finish	M07
262	T079	J31	0	1250	217	164	start	M07
268	T006	J03	4	1250	466	266	finish	M02
268	T039	J15	0	625	219	186	finish	M07
268	T080	J31	0	1250	217	201	start	M02
268	T081	J31	0	1250	217	247	start	M07
271	T028	J07	4	625	311	240	finish	M03
271	T040	J15	0	625	219	174	finish	M03
271	T084	J32	1	1562	1553	227	submit	-
271	T084	J32	1	1562	1553	227	start	M03

time	task	job	priority	CPU	memory	run time	event	machine
274	T068	J29	0	1250	197	50	finish	M03
274	T082	J31	0	1250	217	231	start	M03
279	T066	J27	1	1250	1553	52	finish	M07
279	T083	J31	0	1250	217	248	start	M07
280	T041	J15	0	625	219	170	finish	M06
286	T007	J03	4	1250	466	284	finish	M02
287	T053	J17	1	1562	1553	116	finish	M06
308	T025	J04	0	1250	835	306	finish	M06
310	T022	J04	0	1250	835	308	finish	M05
311	T001	J01	0	736	317	311	finish	M01
311	T064	J25	1	1562	1553	97	finish	M04
318	T085	J33	1	1250	1553	169	submit	-
318	T086	J34	0	1250	16	27	submit	-
318	T085	J33	1	1250	1553	169	start	M01
318	T086	J34	0	1250	16	27	start	M02
320	T043	J15	0	625	219	175	finish	M07
321	T042	J15	0	625	219	176	finish	M04
321	T087	J35	0	1250	197	152	submit	-
321	T088	J35	0	1250	197	106	submit	-
321	T089	J35	0	1250	197	127	submit	-
321	T087	J35	0	1250	197	152	start	M04
321	T088	J35	0	1250	197	106	start	M04
321	T089	J35	0	1250	197	127	start	M05
331	T044	J15	0	625	219	186	finish	M07
331	T045	J15	0	625	219	182	finish	M05
339	T049	J15	0	625	219	170	finish	M03
345	T086	J34	0	1250	16	27	finish	M02
349	T090	J36	1	1562	1553	633	submit	-
349	T090	J36	1	1562	1553	633	start	M02
351	T046	J15	0	625	219	182	finish	M03
352	T047	J15	0	625	219	183	finish	M03
352	T048	J15	0	625	219	183	finish	M03
354	T091	J37	4	1477	197	297	submit	-
354	T092	J37	4	1477	197	282	submit	-
354	T093	J37	4	1477	197	278	submit	-
354	T094	J37	4	1477	197	278	submit	-
354	T091	J37	4	1477	197	297	start	M03
354	T092	J37	4	1477	197	282	start	M05
354	T093	J37	4	1477	197	278	start	M06
354	T094	J37	4	1477	197	278	start	M06
359	T095	J38	9	6250	1590	86	submit	-
359	T077	J31	0	1250	217	183	evict	M01
359	T075	J31	0	1250	217	246	evict	M01
359	T073	J31	0	1250	217	168	evict	M01
359	T072	J31	0	1250	217	188	evict	M01
359	T085	J33	1	1250	1553	169	evict	M01
359	T095	J38	9	6250	1590	86	start	M01
359	T085	J33	1	1250	1553	169	start	M07
359	T072	J31	0	1250	217	188	start	M03
364	T096	J39	1	1562	1553	247	submit	-
368	T050	J15	0	625	219	170	finish	M04
372	T097	J40	0	1250	610	1134	submit	-
372	T098	J40	0	1250	610	1124	submit	-
372	T099	J40	0	1250	610	1125	submit	-
372	T100	J40	0	1250	610	1122	submit	-
372	T101	J40	0	1250	610	1122	submit	-
372	T102	J40	0	1250	610	1121	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
372	T103	J40	0	1250	610	1122	submit	-
372	T104	J40	0	1250	610	1123	submit	-
372	T105	J40	0	1250	610	1122	submit	-
372	T106	J40	0	1250	610	1121	submit	-
372	T107	J40	0	1250	610	1123	submit	-
372	T108	J40	0	1250	610	81	submit	-
376	T051	J15	0	625	219	174	finish	M03
377	T070	J31	0	1250	217	147	finish	M04
377	T096	J39	1	1562	1553	247	start	M04
392	T109	J41	8	3125	3180	149	submit	-
392	T110	J42	8	625	155	165	submit	-
392	T110	J42	8	625	155	165	start	M03
409	T111	J43	1	1250	1553	557	submit	-
426	T061	J22	6	937	1244	277	finish	M05
426	T079	J31	0	1250	217	164	finish	M07
426	T073	J31	0	1250	217	168	start	M07
427	T088	J35	0	1250	197	106	finish	M04
427	T111	J43	1	1250	1553	557	start	M04
434	T078	J31	0	1250	217	178	finish	M06
445	T095	J38	9	6250	1590	86	finish	M01
445	T075	J31	0	1250	217	246	start	M01
445	T077	J31	0	1250	217	183	start	M01
445	T097	J40	0	1250	610	1134	start	M01
445	T098	J40	0	1250	610	1124	start	M01
445	T099	J40	0	1250	610	1125	start	M01
448	T089	J35	0	1250	197	127	finish	M05
448	T100	J40	0	1250	610	1122	start	M05
449	T074	J31	0	1250	217	205	finish	M07
449	T101	J40	0	1250	610	1122	start	M07
457	T027	J06	1	1250	1590	432	finish	M06
457	T109	J41	8	3125	3180	149	start	M06
458	T112	J44	8	937	466	70	submit	-
458	T112	J44	8	937	466	70	start	M05
459	T071	J31	0	1250	217	223	finish	M05
459	T076	J31	0	1250	217	211	finish	M02
459	T102	J40	0	1250	610	1121	start	M02
459	T103	J40	0	1250	610	1122	start	M05
469	T080	J31	0	1250	217	201	finish	M02
473	T087	J35	0	1250	197	152	finish	M04
473	T104	J40	0	1250	610	1123	start	M04
475	T113	J45	4	625	197	53	submit	-
475	T113	J45	4	625	197	53	start	M02
478	T065	J26	1	1250	1553	256	finish	M05
478	T067	J28	0	905	227	254	finish	M02
478	T105	J40	0	1250	610	1122	start	M05
480	T069	J30	0	1250	283	252	finish	M02
480	T106	J40	0	1250	610	1121	start	M02
492	T114	J46	4	1250	466	287	submit	-
498	T084	J32	1	1562	1553	227	finish	M03
498	T114	J46	4	1250	466	287	start	M03
505	T082	J31	0	1250	217	231	finish	M03
505	T107	J40	0	1250	610	1123	start	M03
509	T115	J47	1	1562	1553	121	submit	-
510	T116	J48	6	629	197	74	submit	-
510	T117	J48	6	629	197	71	submit	-
510	T118	J49	4	3125	1125	38	submit	-
515	T081	J31	0	1250	217	247	finish	M07

time	task	job	priority	CPU	memory	run time	event	machine
515	T116	J48	6	629	197	74	start	M07
526	T063	J24	1	1562	1553	325	finish	M04
526	T117	J48	6	629	197	71	start	M04
526	T108	J40	0	1250	610	81	start	M04
527	T083	J31	0	1250	217	248	finish	M07
528	T085	J33	1	1250	1553	169	finish	M07
528	T112	J44	8	937	466	70	finish	M05
528	T113	J45	4	625	197	53	finish	M02
528	T119	J50	1	1562	1553	249	submit	-
547	T072	J31	0	1250	217	188	finish	M03
547	T115	J47	1	1562	1553	121	start	M03
556	T120	J51	1	1250	1553	205	submit	-
556	T121	J52	1	1250	1553	57	submit	-
557	T110	J42	8	625	155	165	finish	M03
558	T122	J53	0	1250	197	55	submit	-
558	T123	J54	0	3125	311	617	submit	-
558	T122	J53	0	1250	197	55	start	M02
585	T092	J37	4	1477	197	282	fail	M05
585	T100	J40	0	1250	610	1122	fail	M05
585	T103	J40	0	1250	610	1122	fail	M05
585	T105	J40	0	1250	610	1122	fail	M05
585	T092	J37	4	1477	197	282	start	M07
589	T116	J48	6	629	197	74	finish	M07
594	T073	J31	0	1250	217	168	finish	M07
594	T118	J49	4	3125	1125	38	start	M07
597	T117	J48	6	629	197	71	finish	M04
606	T109	J41	8	3125	3180	149	finish	M06
606	T119	J50	1	1562	1553	249	start	M06
606	T120	J51	1	1250	1553	205	start	M06
607	T108	J40	0	1250	610	81	finish	M04
607	T121	J52	1	1250	1553	57	start	M04
613	T122	J53	0	1250	197	55	finish	M02
614	T124	J55	4	3125	777	238	submit	-
619	T125	J56	0	6250	466	41	submit	-
619	T126	J57	9	6250	3180	210	submit	-
619	T104	J40	0	1250	610	1123	evict	M04
619	T121	J52	1	1250	1553	57	evict	M04
619	T111	J43	1	1250	1553	557	evict	M04
619	T096	J39	1	1562	1553	247	evict	M04
619	T126	J57	9	6250	3180	210	start	M04
628	T077	J31	0	1250	217	183	finish	M01
628	T100	J40	0	1250	610	1122	start	M01
632	T093	J37	4	1477	197	278	finish	M06
632	T094	J37	4	1477	197	278	finish	M06
632	T118	J49	4	3125	1125	38	finish	M07
632	T124	J55	4	3125	777	238	start	M06
632	T096	J39	1	1562	1553	247	start	M07
632	T103	J40	0	1250	610	1122	start	M07
633	T127	J58	4	625	311	35	submit	-
633	T127	J58	4	625	311	35	start	M02
651	T091	J37	4	1477	197	297	finish	M03
668	T115	J47	1	1562	1553	121	finish	M03
668	T127	J58	4	625	311	35	finish	M02
668	T111	J43	1	1250	1553	557	start	M03
691	T075	J31	0	1250	217	246	finish	M01
691	T104	J40	0	1250	610	1123	start	M01
740	T128	J59	0	6250	1244	77	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
740	T129	J60	0	6250	1244	112	submit	-
754	T130	J61	4	1250	466	260	submit	-
754	T131	J61	4	1250	466	266	submit	-
754	T132	J61	4	1250	466	305	submit	-
754	T133	J61	4	1250	466	302	submit	-
754	T130	J61	4	1250	466	260	start	M03
772	T134	J62	4	1250	466	237	submit	-
772	T135	J62	4	1250	466	275	submit	-
772	T136	J62	4	1250	466	238	submit	-
772	T137	J62	4	1250	466	237	submit	-
785	T114	J46	4	1250	466	287	finish	M03
785	T131	J61	4	1250	466	266	start	M03
811	T120	J51	1	1250	1553	205	finish	M06
811	T132	J61	4	1250	466	305	start	M06
829	T126	J57	9	6250	3180	210	finish	M04
829	T133	J61	4	1250	466	302	start	M04
829	T134	J62	4	1250	466	237	start	M04
829	T135	J62	4	1250	466	275	start	M04
829	T136	J62	4	1250	466	238	start	M04
829	T137	J62	4	1250	466	237	start	M04
855	T119	J50	1	1562	1553	249	finish	M06
855	T121	J52	1	1250	1553	57	start	M06
867	T092	J37	4	1477	197	282	finish	M07
870	T124	J55	4	3125	777	238	finish	M06
870	T105	J40	0	1250	610	1122	start	M06
872	T138	J63	4	3125	1125	101	submit	-
879	T096	J39	1	1562	1553	247	finish	M07
879	T138	J63	4	3125	1125	101	start	M07
912	T121	J52	1	1250	1553	57	finish	M06
912	T123	J54	0	3125	311	617	start	M06
919	T139	J64	4	1250	466	283	submit	-
919	T140	J65	1	5627	199	48	submit	-
938	T141	J66	4	1250	398	43	submit	-
980	T138	J63	4	3125	1125	101	finish	M07
980	T139	J64	4	1250	466	283	start	M07
980	T141	J66	4	1250	398	43	start	M07
982	T090	J36	1	1562	1553	633	finish	M02
1014	T130	J61	4	1250	466	260	finish	M03
1023	T141	J66	4	1250	398	43	finish	M07
1051	T131	J61	4	1250	466	266	finish	M03
1066	T134	J62	4	1250	466	237	finish	M04
1066	T137	J62	4	1250	466	237	finish	M04
1067	T136	J62	4	1250	466	238	finish	M04
1080	T142	J67	9	6250	1590	230	submit	-
1080	T104	J40	0	1250	610	1123	evict	M01
1080	T100	J40	0	1250	610	1122	evict	M01
1080	T099	J40	0	1250	610	1125	evict	M01
1080	T098	J40	0	1250	610	1124	evict	M01
1080	T097	J40	0	1250	610	1134	evict	M01
1080	T142	J67	9	6250	1590	230	start	M01
1080	T097	J40	0	1250	610	1134	start	M02
1080	T098	J40	0	1250	610	1124	start	M02
1080	T099	J40	0	1250	610	1125	start	M02
1080	T100	J40	0	1250	610	1122	start	M03
1080	T104	J40	0	1250	610	1123	start	M07
1091	T143	J68	1	769	466	276	submit	-
1091	T143	J68	1	769	466	276	start	M07

time	task	job	priority	CPU	memory	run time	event	machine
1093	T144	J69	0	625	155	277	submit	-
1093	T145	J69	0	625	155	249	submit	-
1093	T146	J70	0	625	155	287	submit	-
1093	T147	J70	0	625	155	273	submit	-
1093	T148	J71	0	625	155	274	submit	-
1093	T149	J71	0	625	155	269	submit	-
1093	T144	J69	0	625	155	277	start	M03
1093	T145	J69	0	625	155	249	start	M03
1093	T146	J70	0	625	155	287	start	M06
1095	T150	J72	0	625	155	247	submit	-
1095	T151	J72	0	625	155	246	submit	-
1095	T152	J72	0	625	155	248	submit	-
1095	T153	J72	0	625	155	260	submit	-
1095	T154	J72	0	625	155	244	submit	-
1095	T155	J72	0	625	155	265	submit	-
1095	T156	J72	0	625	155	259	submit	-
1095	T157	J72	0	625	155	245	submit	-
1095	T158	J72	0	625	155	268	submit	-
1095	T159	J72	0	625	155	255	submit	-
1095	T160	J72	0	625	155	245	submit	-
1095	T161	J72	0	625	155	245	submit	-
1095	T162	J72	0	625	155	265	submit	-
1095	T163	J72	0	625	155	256	submit	-
1095	T164	J72	0	625	155	243	submit	-
1095	T165	J72	0	625	155	258	submit	-
1095	T166	J73	0	625	155	257	submit	-
1095	T167	J73	0	625	155	259	submit	-
1095	T168	J73	0	625	155	248	submit	-
1095	T169	J73	0	625	155	276	submit	-
1095	T170	J73	0	625	155	283	submit	-
1095	T171	J73	0	625	155	250	submit	-
1095	T172	J73	0	625	155	252	submit	-
1095	T173	J73	0	625	155	258	submit	-
1095	T174	J73	0	625	155	249	submit	-
1095	T175	J73	0	625	155	250	submit	-
1095	T176	J73	0	625	155	280	submit	-
1095	T177	J73	0	625	155	259	submit	-
1095	T178	J73	0	625	155	248	submit	-
1095	T179	J73	0	625	155	261	submit	-
1095	T180	J73	0	625	155	252	submit	-
1095	T181	J73	0	625	155	271	submit	-
1104	T135	J62	4	1250	466	275	finish	M04
1116	T132	J61	4	1250	466	305	finish	M06
1116	T147	J70	0	625	155	273	start	M06
1116	T148	J71	0	625	155	274	start	M06
1131	T133	J61	4	1250	466	302	finish	M04
1131	T140	J65	1	5627	199	48	start	M04
1160	T182	J74	1	625	311	39	submit	-
1174	T183	J75	1	937	777	58	submit	-
1179	T140	J65	1	5627	199	48	finish	M04
1179	T182	J74	1	625	311	39	start	M04
1179	T183	J75	1	937	777	58	start	M04

Cell window 14: Events

Time 359: task T095 (priority 9) evicted task T072 (priority 0) from machine M01.

Time 359: task T095 (priority 9) evicted task T073 (priority 0) from machine M01.

Time 359: task T095 (priority 9) evicted task T075 (priority 0) from machine M01.

Time 359: task T095 (priority 9) evicted task T077 (priority 0) from machine M01.

Time 359: task T095 (priority 9) evicted task T085 (priority 1) from machine M01.

Time 359: task T072 restarted on machine M03.

Time 359: task T085 restarted on machine M07.

Time 426: task T073 restarted on machine M07.

Time 445: task T075 restarted on machine M01.

Time 445: task T077 restarted on machine M01.

Time 585: machine M05 failed; tasks T092, T100, T103, T105 stopped and went back to the queue.

Time 585: task T092 restarted on machine M07.

Time 619: task T126 (priority 9) evicted task T096 (priority 1) from machine M04.

Time 619: task T126 (priority 9) evicted task T104 (priority 0) from machine M04.

Time 619: task T126 (priority 9) evicted task T111 (priority 1) from machine M04.

Time 619: task T126 (priority 9) evicted task T121 (priority 1) from machine M04.

Time 628: task T100 restarted on machine M01.

Time 632: task T096 restarted on machine M07.

Time 632: task T103 restarted on machine M07.

Time 668: task T111 restarted on machine M03.

Time 691: task T104 restarted on machine M01.

Time 855: task T121 restarted on machine M06.

Time 870: task T105 restarted on machine M06.

Time 1080: task T142 (priority 9) evicted task T097 (priority 0) from machine M01.

Time 1080: task T142 (priority 9) evicted task T098 (priority 0) from machine M01.

Time 1080: task T142 (priority 9) evicted task T099 (priority 0) from machine M01.

Time 1080: task T142 (priority 9) evicted task T100 (priority 0) from machine M01.

Time 1080: task T142 (priority 9) evicted task T104 (priority 0) from machine M01.

Time 1080: task T097 restarted on machine M02.

Time 1080: task T098 restarted on machine M02.

Time 1080: task T099 restarted on machine M02.

Time 1080: task T100 restarted on machine M03.

Time 1080: task T104 restarted on machine M07.

Longest waits (submission to first start, tasks started by the window end): task T123 waited 354 s (submitted at 558, first started at 912); task T140 waited 212 s (submitted at 919, first started at 1131); task T108 waited 154 s (submitted at 372, first started at 526); task T051 waited 137 s (submitted at 65, first started at 202); task T050 waited 133 s (submitted at 65, first started at 198).

Cell window 14: Queue at window end

State after the events at time 1200.

Tasks running at time 1200 (remaining run time = seconds still to run after time 1200):

task	job	priority	CPU	memory	machine	started	remaining
T142	J67	9	6250	1590	M01	1080	110
T097	J40	0	1250	610	M02	1080	1014
T098	J40	0	1250	610	M02	1080	1004
T099	J40	0	1250	610	M02	1080	1005
T102	J40	0	1250	610	M02	459	380
T106	J40	0	1250	610	M02	480	401
T100	J40	0	1250	610	M03	1080	1002
T107	J40	0	1250	610	M03	505	428
T111	J43	1	1250	1553	M03	668	25
T144	J69	0	625	155	M03	1093	170
T145	J69	0	625	155	M03	1093	142
T182	J74	1	625	311	M04	1179	18
T183	J75	1	937	777	M04	1179	37
T105	J40	0	1250	610	M06	870	792
T123	J54	0	3125	311	M06	912	329
T146	J70	0	625	155	M06	1093	180
T147	J70	0	625	155	M06	1116	189
T148	J71	0	625	155	M06	1116	190
T101	J40	0	1250	610	M07	449	371
T103	J40	0	1250	610	M07	632	554
T104	J40	0	1250	610	M07	1080	1003
T139	J64	4	1250	466	M07	980	63
T143	J68	1	769	466	M07	1091	167
T033	J12	8	6250	1590	M08	49	127

Tasks waiting at time 1200, in queue order (each will run its full run time once started):

task	job	priority	CPU	memory	submitted	run time
T125	J56	0	6250	466	619	41
T128	J59	0	6250	1244	740	77
T129	J60	0	6250	1244	740	112
T149	J71	0	625	155	1093	269
T150	J72	0	625	155	1095	247
T151	J72	0	625	155	1095	246
T152	J72	0	625	155	1095	248
T153	J72	0	625	155	1095	260
T154	J72	0	625	155	1095	244
T155	J72	0	625	155	1095	265
T156	J72	0	625	155	1095	259
T157	J72	0	625	155	1095	245
T158	J72	0	625	155	1095	268
T159	J72	0	625	155	1095	255
T160	J72	0	625	155	1095	245
T161	J72	0	625	155	1095	245
T162	J72	0	625	155	1095	265
T163	J72	0	625	155	1095	256
T164	J72	0	625	155	1095	243
T165	J72	0	625	155	1095	258
T166	J73	0	625	155	1095	257
T167	J73	0	625	155	1095	259
T168	J73	0	625	155	1095	248
T169	J73	0	625	155	1095	276
T170	J73	0	625	155	1095	283
T171	J73	0	625	155	1095	250
T172	J73	0	625	155	1095	252
T173	J73	0	625	155	1095	258

task	job	priority	CPU	memory	submitted	run time
T174	J73	0	625	155	1095	249
T175	J73	0	625	155	1095	250
T176	J73	0	625	155	1095	280
T177	J73	0	625	155	1095	259
T178	J73	0	625	155	1095	248
T179	J73	0	625	155	1095	261
T180	J73	0	625	155	1095	252
T181	J73	0	625	155	1095	271

Machine M05 is down at time 1200; it returns at time 7370.

Cell window 19

Times 0 to 1200. The cluster has 8 machines. 114 tasks of 35 jobs are submitted during the window.

Cell window 19: Machines

machine	CPU	memory	fails at	returns at
M01	6250	3116	-	-
M02	6250	6244	-	-
M03	6250	3116	-	-
M04	6250	6244	-	-
M05	6250	6244	-	-
M06	6250	6244	525	1435
M07	6250	6244	-	-
M08	6250	3116	-	-

Cell window 19: Task log

time	task	job	priority	CPU	memory	run time	event	machine
46	T001	J01	6	937	1244	276	submit	-
46	T001	J01	6	937	1244	276	start	M01
165	T002	J02	9	6250	233	105	submit	-
165	T002	J02	9	6250	233	105	start	M02
167	T003	J03	1	1250	1590	389	submit	-
167	T003	J03	1	1250	1590	389	start	M01
202	T004	J04	1	1250	466	176	submit	-
202	T004	J04	1	1250	466	176	start	M03
204	T005	J05	0	625	252	173	submit	-
204	T006	J05	0	625	252	173	submit	-
204	T007	J05	0	625	252	173	submit	-
204	T008	J05	0	625	252	173	submit	-
204	T009	J05	0	625	252	173	submit	-
204	T010	J05	0	625	252	172	submit	-
204	T011	J05	0	625	252	173	submit	-
204	T005	J05	0	625	252	173	start	M01
204	T006	J05	0	625	252	173	start	M03
204	T007	J05	0	625	252	173	start	M03
204	T008	J05	0	625	252	173	start	M03
204	T009	J05	0	625	252	173	start	M03
204	T010	J05	0	625	252	172	start	M03
204	T011	J05	0	625	252	173	start	M03
227	T012	J06	0	6250	1244	422	submit	-
227	T012	J06	0	6250	1244	422	start	M04
270	T002	J02	9	6250	233	105	finish	M02
286	T013	J07	4	625	1398	158	submit	-
286	T014	J07	4	625	1398	172	submit	-
286	T015	J07	4	625	1398	162	submit	-
286	T016	J07	4	625	1398	164	submit	-
286	T013	J07	4	625	1398	158	start	M02
286	T014	J07	4	625	1398	172	start	M02
286	T015	J07	4	625	1398	162	start	M02
286	T016	J07	4	625	1398	164	start	M02
322	T001	J01	6	937	1244	276	finish	M01
376	T010	J05	0	625	252	172	finish	M03
377	T005	J05	0	625	252	173	finish	M01
377	T006	J05	0	625	252	173	finish	M03
377	T007	J05	0	625	252	173	finish	M03

time	task	job	priority	CPU	memory	run time	event	machine
377	T008	J05	0	625	252	173	finish	M03
377	T009	J05	0	625	252	173	finish	M03
377	T011	J05	0	625	252	173	finish	M03
378	T004	J04	1	1250	466	176	finish	M03
381	T017	J08	4	4834	3876	499	submit	-
381	T018	J08	4	4834	3876	330	submit	-
381	T019	J08	4	4834	3876	351	submit	-
381	T020	J08	4	4834	3876	316	submit	-
381	T017	J08	4	4834	3876	499	start	M05
381	T018	J08	4	4834	3876	330	start	M06
381	T019	J08	4	4834	3876	351	start	M07
385	T021	J09	1	1562	1553	1107	submit	-
385	T021	J09	1	1562	1553	1107	start	M03
396	T022	J10	4	1874	1590	635	submit	-
396	T023	J10	4	1874	1590	631	submit	-
396	T024	J10	4	1874	1590	637	submit	-
396	T025	J10	4	1874	1590	664	submit	-
396	T026	J11	6	625	197	80	submit	-
396	T026	J11	6	625	197	80	start	M01
396	T022	J10	4	1874	1590	635	start	M08
430	T027	J12	1	1843	1553	1046	submit	-
430	T027	J12	1	1843	1553	1046	start	M03
432	T028	J13	0	1250	822	773	submit	-
432	T029	J13	0	1250	822	813	submit	-
432	T030	J13	0	1250	822	841	submit	-
432	T031	J13	0	1250	822	740	submit	-
432	T032	J13	0	1250	822	720	submit	-
432	T033	J13	0	1250	822	743	submit	-
432	T034	J13	0	1250	822	723	submit	-
432	T035	J13	0	1250	822	753	submit	-
432	T036	J13	0	1250	822	769	submit	-
432	T037	J13	0	1250	822	731	submit	-
432	T038	J13	0	1250	822	792	submit	-
432	T039	J13	0	1250	822	713	submit	-
432	T040	J13	0	1250	822	719	submit	-
432	T041	J13	0	1250	822	713	submit	-
432	T042	J13	0	1250	822	745	submit	-
432	T043	J13	0	1250	822	730	submit	-
432	T028	J13	0	1250	822	773	start	M01
432	T029	J13	0	1250	822	813	start	M05
432	T030	J13	0	1250	822	841	start	M06
432	T031	J13	0	1250	822	740	start	M07
432	T032	J13	0	1250	822	720	start	M08
444	T013	J07	4	625	1398	158	finish	M02
448	T015	J07	4	625	1398	162	finish	M02
450	T016	J07	4	625	1398	164	finish	M02
450	T020	J08	4	4834	3876	316	start	M02
458	T014	J07	4	625	1398	172	finish	M02
458	T033	J13	0	1250	822	743	start	M02
476	T026	J11	6	625	197	80	finish	M01
525	T018	J08	4	4834	3876	330	fail	M06
525	T030	J13	0	1250	822	841	fail	M06
556	T003	J03	1	1250	1590	389	finish	M01
556	T023	J10	4	1874	1590	631	start	M01
598	T044	J14	8	625	466	414	submit	-
598	T044	J14	8	625	466	414	start	M01
649	T012	J06	0	6250	1244	422	finish	M04

time	task	job	priority	CPU	memory	run time	event	machine
649	T018	J08	4	4834	3876	330	start	M04
649	T030	J13	0	1250	822	841	start	M04
650	T045	J15	9	1250	1590	53	submit	-
650	T033	J13	0	1250	822	743	evict	M02
650	T045	J15	9	1250	1590	53	start	M02
654	T046	J16	0	937	266	58	submit	-
654	T047	J16	0	937	266	50	submit	-
654	T048	J16	0	937	266	54	submit	-
654	T049	J16	0	937	266	49	submit	-
654	T050	J16	0	937	266	58	submit	-
654	T051	J16	0	937	266	49	submit	-
654	T052	J16	0	937	266	52	submit	-
654	T053	J16	0	937	266	54	submit	-
654	T054	J16	0	937	266	54	submit	-
654	T055	J16	0	937	266	50	submit	-
654	T056	J16	0	937	266	54	submit	-
654	T057	J16	0	937	266	54	submit	-
654	T058	J16	0	937	266	58	submit	-
654	T059	J16	0	937	266	52	submit	-
654	T060	J16	0	937	266	60	submit	-
654	T061	J16	0	937	266	53	submit	-
654	T046	J16	0	937	266	58	start	M08
654	T047	J16	0	937	266	50	start	M08
658	T062	J17	8	1250	932	323	submit	-
687	T063	J18	6	937	197	65	submit	-
687	T064	J18	6	937	197	76	submit	-
687	T065	J18	6	937	197	62	submit	-
687	T066	J18	6	937	197	71	submit	-
687	T063	J18	6	937	197	65	start	M01
696	T067	J19	9	6250	1590	74	submit	-
696	T027	J12	1	1843	1553	1046	evict	M03
696	T021	J09	1	1562	1553	1107	evict	M03
696	T067	J19	9	6250	1590	74	start	M03
703	T045	J15	9	1250	1590	53	finish	M02
703	T062	J17	8	1250	932	323	start	M02
704	T047	J16	0	937	266	50	finish	M08
704	T064	J18	6	937	197	76	start	M08
704	T065	J18	6	937	197	62	start	M08
705	T068	J20	9	62	466	2775	submit	-
705	T068	J20	9	62	466	2775	start	M02
712	T046	J16	0	937	266	58	finish	M08
712	T066	J18	6	937	197	71	start	M08
726	T069	J21	4	3125	777	241	submit	-
726	T070	J21	4	3125	777	241	submit	-
732	T019	J08	4	4834	3876	351	finish	M07
732	T024	J10	4	1874	1590	637	start	M07
732	T025	J10	4	1874	1590	664	start	M07
732	T033	J13	0	1250	822	743	start	M07
738	T071	J22	0	687	932	2637	submit	-
738	T072	J22	0	687	932	2612	submit	-
738	T073	J22	0	687	932	2620	submit	-
738	T074	J22	0	687	932	2614	submit	-
738	T075	J22	0	687	932	2619	submit	-
738	T076	J22	0	687	932	2617	submit	-
738	T077	J22	0	687	932	2624	submit	-
738	T078	J22	0	687	932	2605	submit	-
738	T079	J22	0	687	932	2624	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
738	T080	J22	0	687	932	2613	submit	-
738	T081	J22	0	687	932	2713	submit	-
738	T082	J22	0	687	932	2613	submit	-
738	T083	J22	0	687	932	2620	submit	-
738	T084	J22	0	687	932	2604	submit	-
738	T085	J22	0	687	932	2624	submit	-
738	T086	J22	0	687	932	2625	submit	-
752	T063	J18	6	937	197	65	finish	M01
766	T020	J08	4	4834	3876	316	finish	M02
766	T065	J18	6	937	197	62	finish	M08
766	T069	J21	4	3125	777	241	start	M02
766	T021	J09	1	1562	1553	1107	start	M02
766	T048	J16	0	937	266	54	start	M08
769	T087	J23	0	6250	466	44	submit	-
770	T067	J19	9	6250	1590	74	finish	M03
770	T070	J21	4	3125	777	241	start	M03
770	T027	J12	1	1843	1553	1046	start	M03
770	T049	J16	0	937	266	49	start	M03
780	T064	J18	6	937	197	76	finish	M08
783	T066	J18	6	937	197	71	finish	M08
783	T050	J16	0	937	266	58	start	M08
801	T088	J24	1	937	1590	1173	submit	-
802	T089	J25	4	1250	466	324	submit	-
803	T090	J26	0	1250	919	1171	submit	-
805	T091	J27	0	937	244	81	submit	-
805	T092	J27	0	937	244	72	submit	-
805	T093	J27	0	937	244	77	submit	-
805	T094	J27	0	937	244	77	submit	-
805	T095	J27	0	937	244	82	submit	-
805	T096	J27	0	937	244	72	submit	-
805	T097	J27	0	937	244	90	submit	-
805	T098	J27	0	937	244	72	submit	-
805	T099	J27	0	937	244	77	submit	-
805	T100	J27	0	937	244	81	submit	-
805	T101	J27	0	937	244	77	submit	-
805	T102	J27	0	937	244	78	submit	-
805	T103	J27	0	937	244	70	submit	-
805	T104	J27	0	937	244	108	submit	-
805	T105	J27	0	937	244	89	submit	-
805	T106	J27	0	937	244	73	submit	-
819	T049	J16	0	937	266	49	finish	M03
819	T089	J25	4	1250	466	324	start	M03
820	T048	J16	0	937	266	54	finish	M08
820	T051	J16	0	937	266	49	start	M08
840	T107	J28	8	5627	3180	261	submit	-
841	T050	J16	0	937	266	58	finish	M08
841	T052	J16	0	937	266	52	start	M08
869	T051	J16	0	937	266	49	finish	M08
869	T053	J16	0	937	266	54	start	M08
880	T017	J08	4	4834	3876	499	finish	M05
880	T054	J16	0	937	266	54	start	M05
880	T055	J16	0	937	266	50	start	M05
880	T056	J16	0	937	266	54	start	M05
880	T057	J16	0	937	266	54	start	M05
880	T058	J16	0	937	266	58	start	M05
885	T108	J29	0	6250	1244	58	submit	-
888	T109	J30	9	6250	3528	2929	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
888	T058	J16	0	937	266	58	evict	M05
888	T057	J16	0	937	266	54	evict	M05
888	T056	J16	0	937	266	54	evict	M05
888	T055	J16	0	937	266	50	evict	M05
888	T054	J16	0	937	266	54	evict	M05
888	T029	J13	0	1250	822	813	evict	M05
888	T109	J30	9	6250	3528	2929	start	M05
893	T052	J16	0	937	266	52	finish	M08
893	T054	J16	0	937	266	54	start	M08
923	T053	J16	0	937	266	54	finish	M08
923	T055	J16	0	937	266	50	start	M08
947	T054	J16	0	937	266	54	finish	M08
947	T056	J16	0	937	266	54	start	M08
973	T055	J16	0	937	266	50	finish	M08
973	T057	J16	0	937	266	54	start	M08
979	T018	J08	4	4834	3876	330	finish	M04
979	T088	J24	1	937	1590	1173	start	M04
979	T029	J13	0	1250	822	813	start	M04
979	T034	J13	0	1250	822	723	start	M04
979	T035	J13	0	1250	822	753	start	M04
991	T110	J31	8	625	398	558	submit	-
1001	T056	J16	0	937	266	54	finish	M08
1001	T110	J31	8	625	398	558	start	M08
1007	T069	J21	4	3125	777	241	finish	M02
1007	T036	J13	0	1250	822	769	start	M02
1007	T037	J13	0	1250	822	731	start	M02
1007	T071	J22	0	687	932	2637	start	M02
1011	T070	J21	4	3125	777	241	finish	M03
1011	T038	J13	0	1250	822	792	start	M03
1011	T058	J16	0	937	266	58	start	M03
1012	T044	J14	8	625	466	414	finish	M01
1012	T059	J16	0	937	266	52	start	M01
1012	T060	J16	0	937	266	60	start	M01
1016	T111	J32	8	6250	777	424	submit	-
1016	T112	J33	8	6250	777	340	submit	-
1026	T062	J17	8	1250	932	323	finish	M02
1026	T039	J13	0	1250	822	713	start	M02
1027	T057	J16	0	937	266	54	finish	M08
1027	T061	J16	0	937	266	53	start	M08
1031	T022	J10	4	1874	1590	635	finish	M08
1031	T040	J13	0	1250	822	719	start	M08
1031	T091	J27	0	937	244	81	start	M08
1031	T092	J27	0	937	244	72	start	M08
1064	T059	J16	0	937	266	52	finish	M01
1064	T093	J27	0	937	244	77	start	M01
1066	T113	J34	9	6250	233	114	submit	-
1066	T035	J13	0	1250	822	753	evict	M04
1066	T034	J13	0	1250	822	723	evict	M04
1066	T029	J13	0	1250	822	813	evict	M04
1066	T030	J13	0	1250	822	841	evict	M04
1066	T088	J24	1	937	1590	1173	evict	M04
1066	T113	J34	9	6250	233	114	start	M04
1069	T058	J16	0	937	266	58	finish	M03
1069	T094	J27	0	937	244	77	start	M03
1072	T060	J16	0	937	266	60	finish	M01
1072	T095	J27	0	937	244	82	start	M01
1080	T061	J16	0	937	266	53	finish	M08

time	task	job	priority	CPU	memory	run time	event	machine
1080	T096	J27	0	937	244	72	start	M08
1096	T114	J35	8	6250	1166	243	submit	-
1103	T092	J27	0	937	244	72	finish	M08
1103	T097	J27	0	937	244	90	start	M08
1112	T091	J27	0	937	244	81	finish	M08
1112	T098	J27	0	937	244	72	start	M08
1141	T093	J27	0	937	244	77	finish	M01
1141	T099	J27	0	937	244	77	start	M01
1143	T089	J25	4	1250	466	324	finish	M03
1143	T100	J27	0	937	244	81	start	M03
1143	T101	J27	0	937	244	77	start	M03
1146	T094	J27	0	937	244	77	finish	M03
1146	T102	J27	0	937	244	78	start	M03
1152	T032	J13	0	1250	822	720	finish	M08
1152	T096	J27	0	937	244	72	finish	M08
1152	T029	J13	0	1250	822	813	start	M08
1152	T103	J27	0	937	244	70	start	M08
1154	T095	J27	0	937	244	82	finish	M01
1154	T104	J27	0	937	244	108	start	M01
1172	T031	J13	0	1250	822	740	finish	M07
1172	T088	J24	1	937	1590	1173	start	M07
1180	T113	J34	9	6250	233	114	finish	M04
1180	T107	J28	8	5627	3180	261	start	M04
1184	T098	J27	0	937	244	72	finish	M08
1184	T105	J27	0	937	244	89	start	M08
1187	T023	J10	4	1874	1590	631	finish	M01
1187	T106	J27	0	937	244	73	start	M01
1193	T097	J27	0	937	244	90	finish	M08

Cell window 19: Events

Time 525: machine M06 failed; tasks T018, T030 stopped and went back to the queue.

Time 649: task T018 restarted on machine M04.

Time 649: task T030 restarted on machine M04.

Time 650: task T045 (priority 9) evicted task T033 (priority 0) from machine M02.

Time 696: task T067 (priority 9) evicted task T021 (priority 1) from machine M03.

Time 696: task T067 (priority 9) evicted task T027 (priority 1) from machine M03.

Time 732: task T033 restarted on machine M07.

Time 766: task T021 restarted on machine M02.

Time 770: task T027 restarted on machine M03.

Time 888: task T109 (priority 9) evicted task T029 (priority 0) from machine M05.

Time 888: task T109 (priority 9) evicted task T054 (priority 0) from machine M05.

Time 888: task T109 (priority 9) evicted task T055 (priority 0) from machine M05.

Time 888: task T109 (priority 9) evicted task T056 (priority 0) from machine M05.

Time 888: task T109 (priority 9) evicted task T057 (priority 0) from machine M05.

Time 888: task T109 (priority 9) evicted task T058 (priority 0) from machine M05.

Time 893: task T054 restarted on machine M08.

Time 923: task T055 restarted on machine M08.

Time 947: task T056 restarted on machine M08.

Time 973: task T057 restarted on machine M08.

Time 979: task T029 restarted on machine M04.

Time 1011: task T058 restarted on machine M03.

Time 1066: task T113 (priority 9) evicted task T029 (priority 0) from machine M04.

Time 1066: task T113 (priority 9) evicted task T030 (priority 0) from machine M04.

Time 1066: task T113 (priority 9) evicted task T034 (priority 0) from machine M04.

Time 1066: task T113 (priority 9) evicted task T035 (priority 0) from machine M04.

Time 1066: task T113 (priority 9) evicted task T088 (priority 1) from machine M04.

Time 1152: task T029 restarted on machine M08.

Time 1172: task T088 restarted on machine M07.

Longest waits (submission to first start, tasks started by the window end): task T040 waited 599 s (submitted at 432, first started at 1031); task T039 waited 594 s (submitted at 432, first started at 1026); task T038 waited 579 s (submitted at 432, first started at 1011); task T036 waited 575 s (submitted at 432, first started at 1007); task T037 waited 575 s (submitted at 432, first started at 1007).

Cell window 19: Queue at window end

State after the events at time 1200.

Tasks running at time 1200 (remaining run time = seconds still to run after time 1200):

task	job	priority	CPU	memory	machine	started	remaining
T028	J13	0	1250	822	M01	432	5
T099	J27	0	937	244	M01	1141	18
T104	J27	0	937	244	M01	1154	62
T106	J27	0	937	244	M01	1187	60
T021	J09	1	1562	1553	M02	766	673
T036	J13	0	1250	822	M02	1007	576
T037	J13	0	1250	822	M02	1007	538
T039	J13	0	1250	822	M02	1026	539
T068	J20	9	62	466	M02	705	2280
T071	J22	0	687	932	M02	1007	2444
T027	J12	1	1843	1553	M03	770	616
T038	J13	0	1250	822	M03	1011	603
T100	J27	0	937	244	M03	1143	24
T101	J27	0	937	244	M03	1143	20
T102	J27	0	937	244	M03	1146	24
T107	J28	8	5627	3180	M04	1180	241
T109	J30	9	6250	3528	M05	888	2617
T024	J10	4	1874	1590	M07	732	169
T025	J10	4	1874	1590	M07	732	196
T033	J13	0	1250	822	M07	732	275
T088	J24	1	937	1590	M07	1172	1145
T029	J13	0	1250	822	M08	1152	765
T040	J13	0	1250	822	M08	1031	550
T103	J27	0	937	244	M08	1152	22
T105	J27	0	937	244	M08	1184	73
T110	J31	8	625	398	M08	1001	359

Tasks waiting at time 1200, in queue order (each will run its full run time once started):

task	job	priority	CPU	memory	submitted	run time
T111	J32	8	6250	777	1016	424
T112	J33	8	6250	777	1016	340
T114	J35	8	6250	1166	1096	243
T030	J13	0	1250	822	432	841
T034	J13	0	1250	822	432	723
T035	J13	0	1250	822	432	753
T041	J13	0	1250	822	432	713
T042	J13	0	1250	822	432	745
T043	J13	0	1250	822	432	730
T072	J22	0	687	932	738	2612
T073	J22	0	687	932	738	2620
T074	J22	0	687	932	738	2614
T075	J22	0	687	932	738	2619
T076	J22	0	687	932	738	2617
T077	J22	0	687	932	738	2624
T078	J22	0	687	932	738	2605
T079	J22	0	687	932	738	2624
T080	J22	0	687	932	738	2613
T081	J22	0	687	932	738	2713
T082	J22	0	687	932	738	2613
T083	J22	0	687	932	738	2620
T084	J22	0	687	932	738	2604
T085	J22	0	687	932	738	2624
T086	J22	0	687	932	738	2625
T087	J23	0	6250	466	769	44
T090	J26	0	1250	919	803	1171
T108	J29	0	6250	1244	885	58

Machine M06 is down at time 1200; it returns at time 1435.

Cell window 21

Times 0 to 1200. The cluster has 8 machines. 92 tasks of 37 jobs are submitted during the window.

Cell window 21: Machines

machine	CPU	memory	fails at	returns at
M01	6250	3116	-	-
M02	6250	6244	-	-
M03	6250	3116	-	-
M04	6250	6244	-	-
M05	6250	3116	-	-
M06	6250	9363	556	964
M07	6250	6244	-	-
M08	6250	6244	-	-

Cell window 21: Task log

time	task	job	priority	CPU	memory	run time	event	machine
16	T001	J01	9	6250	1590	74	submit	-
16	T001	J01	9	6250	1590	74	start	M01
25	T002	J02	9	62	466	2775	submit	-
25	T002	J02	9	62	466	2775	start	M02
31	T003	J03	8	3125	1590	257	submit	-
31	T003	J03	8	3125	1590	257	start	M02
46	T004	J04	4	3125	777	241	submit	-
46	T005	J04	4	3125	777	241	submit	-
46	T004	J04	4	3125	777	241	start	M03
46	T005	J04	4	3125	777	241	start	M03
58	T006	J05	0	687	932	2637	submit	-
58	T007	J05	0	687	932	2612	submit	-
58	T008	J05	0	687	932	2620	submit	-
58	T009	J05	0	687	932	2614	submit	-
58	T010	J05	0	687	932	2619	submit	-
58	T011	J05	0	687	932	2617	submit	-
58	T012	J05	0	687	932	2624	submit	-
58	T013	J05	0	687	932	2605	submit	-
58	T014	J05	0	687	932	2624	submit	-
58	T015	J05	0	687	932	2613	submit	-
58	T016	J05	0	687	932	2713	submit	-
58	T017	J05	0	687	932	2613	submit	-
58	T018	J05	0	687	932	2620	submit	-
58	T019	J05	0	687	932	2604	submit	-
58	T020	J05	0	687	932	2624	submit	-
58	T021	J05	0	687	932	2625	submit	-
58	T006	J05	0	687	932	2637	start	M02
58	T007	J05	0	687	932	2612	start	M02
58	T008	J05	0	687	932	2620	start	M02
58	T009	J05	0	687	932	2614	start	M02
58	T010	J05	0	687	932	2619	start	M04
58	T011	J05	0	687	932	2617	start	M04
58	T012	J05	0	687	932	2624	start	M04
58	T013	J05	0	687	932	2605	start	M04
58	T014	J05	0	687	932	2624	start	M04
58	T015	J05	0	687	932	2613	start	M04
58	T016	J05	0	687	932	2713	start	M05
58	T017	J05	0	687	932	2613	start	M05

time	task	job	priority	CPU	memory	run time	event	machine
58	T018	J05	0	687	932	2620	start	M05
58	T019	J05	0	687	932	2604	start	M06
58	T020	J05	0	687	932	2624	start	M06
58	T021	J05	0	687	932	2625	start	M06
87	T022	J06	9	6250	3180	236	submit	-
87	T022	J06	9	6250	3180	236	start	M07
90	T001	J01	9	6250	1590	74	finish	M01
108	T023	J07	6	625	3180	28	submit	-
108	T024	J08	8	625	398	326	submit	-
108	T024	J08	8	625	398	326	start	M01
108	T023	J07	6	625	3180	28	start	M06
123	T025	J09	4	1250	466	324	submit	-
123	T026	J09	4	1250	466	311	submit	-
123	T027	J09	4	1250	466	311	submit	-
123	T028	J09	4	1250	466	303	submit	-
123	T029	J10	0	1250	919	1171	submit	-
123	T025	J09	4	1250	466	324	start	M01
123	T026	J09	4	1250	466	311	start	M01
123	T027	J09	4	1250	466	311	start	M01
123	T028	J09	4	1250	466	303	start	M01
123	T029	J10	0	1250	919	1171	start	M06
125	T030	J11	0	937	244	81	submit	-
125	T031	J11	0	937	244	72	submit	-
125	T032	J11	0	937	244	77	submit	-
125	T033	J11	0	937	244	77	submit	-
125	T034	J11	0	937	244	82	submit	-
125	T035	J11	0	937	244	72	submit	-
125	T036	J11	0	937	244	90	submit	-
125	T037	J11	0	937	244	72	submit	-
125	T038	J11	0	937	244	77	submit	-
125	T039	J11	0	937	244	81	submit	-
125	T040	J11	0	937	244	77	submit	-
125	T041	J11	0	937	244	78	submit	-
125	T042	J11	0	937	244	70	submit	-
125	T043	J11	0	937	244	108	submit	-
125	T044	J11	0	937	244	89	submit	-
125	T045	J11	0	937	244	73	submit	-
125	T030	J11	0	937	244	81	start	M04
125	T031	J11	0	937	244	72	start	M04
125	T032	J11	0	937	244	77	start	M05
125	T033	J11	0	937	244	77	start	M06
125	T034	J11	0	937	244	82	start	M06
125	T035	J11	0	937	244	72	start	M08
125	T036	J11	0	937	244	90	start	M08
125	T037	J11	0	937	244	72	start	M08
125	T038	J11	0	937	244	77	start	M08
125	T039	J11	0	937	244	81	start	M08
125	T040	J11	0	937	244	77	start	M08
136	T023	J07	6	625	3180	28	finish	M06
136	T041	J11	0	937	244	78	start	M06
160	T046	J12	8	5627	3180	261	submit	-
176	T047	J13	4	1250	610	209	submit	-
176	T048	J13	4	1250	610	174	submit	-
176	T049	J13	4	1250	610	174	submit	-
176	T050	J13	4	1250	610	174	submit	-
197	T031	J11	0	937	244	72	finish	M04
197	T035	J11	0	937	244	72	finish	M08

time	task	job	priority	CPU	memory	run time	event	machine
197	T037	J11	0	937	244	72	finish	M08
197	T042	J11	0	937	244	70	start	M04
202	T032	J11	0	937	244	77	finish	M05
202	T033	J11	0	937	244	77	finish	M06
202	T038	J11	0	937	244	77	finish	M08
202	T040	J11	0	937	244	77	finish	M08
202	T043	J11	0	937	244	108	start	M05
202	T044	J11	0	937	244	89	start	M06
206	T030	J11	0	937	244	81	finish	M04
206	T039	J11	0	937	244	81	finish	M08
206	T051	J14	9	4688	6360	560	submit	-
206	T044	J11	0	937	244	89	evict	M06
206	T041	J11	0	937	244	78	evict	M06
206	T034	J11	0	937	244	82	evict	M06
206	T029	J10	0	1250	919	1171	evict	M06
206	T021	J05	0	687	932	2625	evict	M06
206	T051	J14	9	4688	6360	560	start	M06
206	T034	J11	0	937	244	82	start	M04
208	T052	J15	9	6250	3528	2929	submit	-
208	T034	J11	0	937	244	82	evict	M04
208	T042	J11	0	937	244	70	evict	M04
208	T015	J05	0	687	932	2613	evict	M04
208	T014	J05	0	687	932	2624	evict	M04
208	T013	J05	0	687	932	2605	evict	M04
208	T012	J05	0	687	932	2624	evict	M04
208	T011	J05	0	687	932	2617	evict	M04
208	T010	J05	0	687	932	2619	evict	M04
208	T052	J15	9	6250	3528	2929	start	M04
215	T036	J11	0	937	244	90	finish	M08
215	T046	J12	8	5627	3180	261	start	M08
230	T053	J16	4	2670	2704	582	submit	-
230	T054	J16	4	2670	2704	592	submit	-
230	T055	J16	4	2670	2704	595	submit	-
230	T056	J16	4	2670	2704	590	submit	-
271	T057	J17	4	3125	1125	32	submit	-
287	T004	J04	4	3125	777	241	finish	M03
287	T005	J04	4	3125	777	241	finish	M03
287	T058	J18	4	1250	155	12	submit	-
287	T047	J13	4	1250	610	209	start	M03
287	T048	J13	4	1250	610	174	start	M03
287	T049	J13	4	1250	610	174	start	M03
287	T050	J13	4	1250	610	174	start	M03
287	T058	J18	4	1250	155	12	start	M03
288	T003	J03	8	3125	1590	257	finish	M02
288	T057	J17	4	3125	1125	32	start	M02
299	T058	J18	4	1250	155	12	finish	M03
299	T034	J11	0	937	244	82	start	M03
310	T043	J11	0	937	244	108	finish	M05
310	T041	J11	0	937	244	78	start	M05
311	T059	J19	8	625	398	558	submit	-
311	T059	J19	8	625	398	558	start	M01
320	T057	J17	4	3125	1125	32	finish	M02
320	T010	J05	0	687	932	2619	start	M02
320	T011	J05	0	687	932	2617	start	M02
323	T022	J06	9	6250	3180	236	finish	M07
323	T053	J16	4	2670	2704	582	start	M07
323	T054	J16	4	2670	2704	592	start	M07

time	task	job	priority	CPU	memory	run time	event	machine
336	T060	J20	8	6250	777	424	submit	-
375	T061	J21	8	1874	1590	481	submit	-
381	T034	J11	0	937	244	82	finish	M03
381	T042	J11	0	937	244	70	start	M03
388	T041	J11	0	937	244	78	finish	M05
388	T044	J11	0	937	244	89	start	M05
402	T062	J22	4	3125	1125	42	submit	-
416	T063	J23	8	6250	1166	243	submit	-
426	T028	J09	4	1250	466	303	finish	M01
426	T029	J10	0	1250	919	1171	start	M01
434	T024	J08	8	625	398	326	finish	M01
434	T026	J09	4	1250	466	311	finish	M01
434	T027	J09	4	1250	466	311	finish	M01
434	T062	J22	4	3125	1125	42	start	M01
447	T025	J09	4	1250	466	324	finish	M01
447	T045	J11	0	937	244	73	start	M01
451	T042	J11	0	937	244	70	finish	M03
456	T064	J24	4	3125	1125	80	submit	-
461	T048	J13	4	1250	610	174	finish	M03
461	T049	J13	4	1250	610	174	finish	M03
461	T050	J13	4	1250	610	174	finish	M03
461	T061	J21	8	1874	1590	481	start	M03
476	T046	J12	8	5627	3180	261	finish	M08
476	T062	J22	4	3125	1125	42	finish	M01
476	T060	J20	8	6250	777	424	start	M08
476	T064	J24	4	3125	1125	80	start	M01
477	T044	J11	0	937	244	89	finish	M05
496	T047	J13	4	1250	610	209	finish	M03
496	T012	J05	0	687	932	2624	start	M03
518	T065	J25	4	1874	1422	243	submit	-
518	T066	J25	4	1874	1422	240	submit	-
518	T067	J25	4	1874	1422	239	submit	-
518	T068	J25	4	1874	1422	259	submit	-
520	T045	J11	0	937	244	73	finish	M01
556	T064	J24	4	3125	1125	80	finish	M01
556	T019	J05	0	687	932	2604	fail	M06
556	T020	J05	0	687	932	2624	fail	M06
556	T051	J14	9	4688	6360	560	fail	M06
556	T065	J25	4	1874	1422	243	start	M01
693	T069	J26	4	625	155	34	submit	-
693	T069	J26	4	625	155	34	start	M01
721	T070	J27	4	1250	466	246	submit	-
721	T071	J27	4	1250	466	243	submit	-
721	T072	J27	4	1250	466	244	submit	-
721	T073	J27	4	1250	466	253	submit	-
721	T070	J27	4	1250	466	246	start	M03
727	T069	J26	4	625	155	34	finish	M01
799	T065	J25	4	1874	1422	243	finish	M01
799	T066	J25	4	1874	1422	240	start	M01
804	T074	J28	8	5627	1590	139	submit	-
840	T075	J29	8	6250	777	216	submit	-
869	T059	J19	8	625	398	558	finish	M01
869	T071	J27	4	1250	466	243	start	M01
877	T076	J30	1	1562	1553	95	submit	-
900	T060	J20	8	6250	777	424	finish	M08
900	T063	J23	8	6250	1166	243	start	M08
905	T053	J16	4	2670	2704	582	finish	M07

time	task	job	priority	CPU	memory	run time	event	machine
915	T054	J16	4	2670	2704	592	finish	M07
915	T074	J28	8	5627	1590	139	start	M07
924	T077	J31	9	4688	6360	535	submit	-
934	T078	J32	8	3125	622	47	submit	-
942	T061	J21	8	1874	1590	481	finish	M03
942	T078	J32	8	3125	622	47	start	M03
942	T013	J05	0	687	932	2605	start	M03
964	T051	J14	9	4688	6360	560	start	M06
964	T072	J27	4	1250	466	244	start	M06
967	T070	J27	4	1250	466	246	finish	M03
967	T073	J27	4	1250	466	253	start	M03
972	T079	J33	4	1250	466	261	submit	-
972	T080	J33	4	1250	466	248	submit	-
972	T081	J33	4	1250	466	246	submit	-
972	T082	J33	4	1250	466	250	submit	-
989	T078	J32	8	3125	622	47	finish	M03
989	T079	J33	4	1250	466	261	start	M03
1035	T083	J34	4	1250	155	21	submit	-
1035	T083	J34	4	1250	155	21	start	M01
1036	T084	J35	4	1250	466	299	submit	-
1036	T085	J35	4	1250	466	289	submit	-
1036	T086	J35	4	1250	466	289	submit	-
1036	T087	J35	4	1250	466	287	submit	-
1039	T066	J25	4	1874	1422	240	finish	M01
1039	T067	J25	4	1874	1422	239	start	M01
1051	T088	J36	4	1250	466	263	submit	-
1051	T089	J36	4	1250	466	226	submit	-
1051	T090	J36	4	1250	466	228	submit	-
1051	T091	J36	4	1250	466	226	submit	-
1054	T074	J28	8	5627	1590	139	finish	M07
1054	T075	J29	8	6250	777	216	start	M07
1056	T083	J34	4	1250	155	21	finish	M01
1088	T092	J37	8	6250	777	146	submit	-
1112	T071	J27	4	1250	466	243	finish	M01
1112	T080	J33	4	1250	466	248	start	M01
1143	T063	J23	8	6250	1166	243	finish	M08
1143	T092	J37	8	6250	777	146	start	M08

Cell window 21: Events

Time 206: task T051 (priority 9) evicted task T021 (priority 0) from machine M06.

Time 206: task T051 (priority 9) evicted task T029 (priority 0) from machine M06.

Time 206: task T051 (priority 9) evicted task T034 (priority 0) from machine M06.

Time 206: task T051 (priority 9) evicted task T041 (priority 0) from machine M06.

Time 206: task T051 (priority 9) evicted task T044 (priority 0) from machine M06.

Time 206: task T034 restarted on machine M04.

Time 208: task T052 (priority 9) evicted task T010 (priority 0) from machine M04.

Time 208: task T052 (priority 9) evicted task T011 (priority 0) from machine M04.

Time 208: task T052 (priority 9) evicted task T012 (priority 0) from machine M04.

Time 208: task T052 (priority 9) evicted task T013 (priority 0) from machine M04.

Time 208: task T052 (priority 9) evicted task T014 (priority 0) from machine M04.

Time 208: task T052 (priority 9) evicted task T015 (priority 0) from machine M04.

Time 208: task T052 (priority 9) evicted task T034 (priority 0) from machine M04.

Time 208: task T052 (priority 9) evicted task T042 (priority 0) from machine M04.

Time 299: task T034 restarted on machine M03.

Time 310: task T041 restarted on machine M05.

Time 320: task T010 restarted on machine M02.

Time 320: task T011 restarted on machine M02.

Time 381: task T042 restarted on machine M03.

Time 388: task T044 restarted on machine M05.

Time 426: task T029 restarted on machine M01.

Time 496: task T012 restarted on machine M03.

Time 556: machine M06 failed; tasks T019, T020, T051 stopped and went back to the queue.

Time 942: task T013 restarted on machine M03.

Time 964: machine M06 returned.

Time 964: task T051 restarted on machine M06.

Longest waits (submission to first start, tasks started by the window end): task T067 waited 521 s (submitted at 518, first started at 1039); task T063 waited 484 s (submitted at 416, first started at 900); task T045 waited 322 s (submitted at 125, first started at 447); task T066 waited 281 s (submitted at 518, first started at 799); task T073 waited 246 s (submitted at 721, first started at 967).

Cell window 21: Queue at window end

State after the events at time 1200.

Tasks running at time 1200 (remaining run time = seconds still to run after time 1200):

task	job	priority	CPU	memory	machine	started	remaining
T029	J10	0	1250	919	M01	426	397
T067	J25	4	1874	1422	M01	1039	78
T080	J33	4	1250	466	M01	1112	160
T002	J02	9	62	466	M02	25	1600
T006	J05	0	687	932	M02	58	1495
T007	J05	0	687	932	M02	58	1470
T008	J05	0	687	932	M02	58	1478
T009	J05	0	687	932	M02	58	1472
T010	J05	0	687	932	M02	320	1739
T011	J05	0	687	932	M02	320	1737
T012	J05	0	687	932	M03	496	1920
T013	J05	0	687	932	M03	942	2347
T073	J27	4	1250	466	M03	967	20
T079	J33	4	1250	466	M03	989	50
T052	J15	9	6250	3528	M04	208	1937
T016	J05	0	687	932	M05	58	1571
T017	J05	0	687	932	M05	58	1471
T018	J05	0	687	932	M05	58	1478
T051	J14	9	4688	6360	M06	964	324
T072	J27	4	1250	466	M06	964	8
T075	J29	8	6250	777	M07	1054	70
T092	J37	8	6250	777	M08	1143	89

Tasks waiting at time 1200, in queue order (each will run its full run time once started):

task	job	priority	CPU	memory	submitted	run time
T077	J31	9	4688	6360	924	535
T055	J16	4	2670	2704	230	595
T056	J16	4	2670	2704	230	590
T068	J25	4	1874	1422	518	259
T081	J33	4	1250	466	972	246
T082	J33	4	1250	466	972	250
T084	J35	4	1250	466	1036	299
T085	J35	4	1250	466	1036	289
T086	J35	4	1250	466	1036	289
T087	J35	4	1250	466	1036	287
T088	J36	4	1250	466	1051	263
T089	J36	4	1250	466	1051	226
T090	J36	4	1250	466	1051	228
T091	J36	4	1250	466	1051	226
T076	J30	1	1562	1553	877	95
T014	J05	0	687	932	58	2624
T015	J05	0	687	932	58	2613
T019	J05	0	687	932	58	2604
T020	J05	0	687	932	58	2624
T021	J05	0	687	932	58	2625

All machines are up at time 1200.

Cell window 24

Times 0 to 1200. The cluster has 8 machines. 212 tasks of 94 jobs are submitted during the window.

Cell window 24: Machines

machine	CPU	memory	fails at	returns at
M01	6250	3116	-	-
M02	6250	6244	-	-
M03	6250	6244	-	-
M04	6250	6244	-	-
M05	6250	6244	338	11189
M06	6250	6244	-	-
M07	6250	6244	-	-
M08	6250	3116	-	-

Cell window 24: Task log

time	task	job	priority	CPU	memory	run time	event	machine
3	T001	J01	0	6250	1244	53	submit	-
3	T002	J02	0	6250	466	44	submit	-
3	T003	J03	0	6250	1244	65	submit	-
3	T001	J01	0	6250	1244	53	start	M01
3	T002	J02	0	6250	466	44	start	M02
3	T003	J03	0	6250	1244	65	start	M03
6	T004	J04	9	6250	1590	71	submit	-
6	T004	J04	9	6250	1590	71	start	M04
7	T005	J05	1	1250	1590	393	submit	-
7	T005	J05	1	1250	1590	393	start	M05
8	T006	J06	6	625	280	266	submit	-
8	T006	J06	6	625	280	266	start	M05
22	T007	J07	4	625	155	26	submit	-
22	T008	J08	6	625	280	120	submit	-
22	T009	J09	4	625	155	33	submit	-
22	T010	J10	4	3125	1125	25	submit	-
22	T011	J11	4	625	311	230	submit	-
22	T008	J08	6	625	280	120	start	M05
22	T007	J07	4	625	155	26	start	M05
22	T009	J09	4	625	155	33	start	M05
22	T010	J10	4	3125	1125	25	start	M06
22	T011	J11	4	625	311	230	start	M05
40	T012	J12	6	1250	586	224	submit	-
40	T013	J12	6	1250	586	182	submit	-
40	T014	J12	6	1250	586	211	submit	-
40	T015	J12	6	1250	586	190	submit	-
40	T016	J13	6	937	229	62	submit	-
40	T017	J13	6	937	229	69	submit	-
40	T018	J13	6	937	229	70	submit	-
40	T019	J13	6	937	229	68	submit	-
40	T012	J12	6	1250	586	224	start	M05
40	T013	J12	6	1250	586	182	start	M06
40	T014	J12	6	1250	586	211	start	M06
40	T015	J12	6	1250	586	190	start	M07
40	T016	J13	6	937	229	62	start	M07
40	T017	J13	6	937	229	69	start	M07
40	T018	J13	6	937	229	70	start	M07
40	T019	J13	6	937	229	68	start	M07

time	task	job	priority	CPU	memory	run time	event	machine
47	T002	J02	0	6250	466	44	finish	M02
47	T010	J10	4	3125	1125	25	finish	M06
48	T007	J07	4	625	155	26	finish	M05
54	T020	J14	1	1250	466	168	submit	-
54	T020	J14	1	1250	466	168	start	M02
55	T009	J09	4	625	155	33	finish	M05
55	T021	J15	6	645	197	75	submit	-
55	T022	J16	6	937	197	75	submit	-
55	T023	J16	6	937	197	63	submit	-
55	T024	J16	6	937	197	64	submit	-
55	T025	J16	6	937	197	60	submit	-
55	T021	J15	6	645	197	75	start	M02
55	T022	J16	6	937	197	75	start	M02
55	T023	J16	6	937	197	63	start	M02
55	T024	J16	6	937	197	64	start	M02
55	T025	J16	6	937	197	60	start	M02
56	T001	J01	0	6250	1244	53	finish	M01
56	T026	J17	0	625	229	166	submit	-
56	T027	J17	0	625	229	156	submit	-
56	T028	J17	0	625	229	166	submit	-
56	T029	J17	0	625	229	155	submit	-
56	T030	J17	0	625	229	156	submit	-
56	T031	J17	0	625	229	156	submit	-
56	T032	J17	0	625	229	156	submit	-
56	T033	J17	0	625	229	156	submit	-
56	T034	J17	0	625	229	157	submit	-
56	T035	J17	0	625	229	154	submit	-
56	T036	J17	0	625	229	155	submit	-
56	T037	J17	0	625	229	156	submit	-
56	T038	J17	0	625	229	155	submit	-
56	T039	J17	0	625	229	166	submit	-
56	T040	J17	0	625	229	155	submit	-
56	T041	J17	0	625	229	154	submit	-
56	T026	J17	0	625	229	166	start	M01
56	T027	J17	0	625	229	156	start	M01
56	T028	J17	0	625	229	166	start	M01
56	T029	J17	0	625	229	155	start	M01
56	T030	J17	0	625	229	156	start	M01
56	T031	J17	0	625	229	156	start	M01
56	T032	J17	0	625	229	156	start	M01
56	T033	J17	0	625	229	156	start	M01
56	T034	J17	0	625	229	157	start	M01
56	T035	J17	0	625	229	154	start	M01
56	T036	J17	0	625	229	155	start	M05
56	T037	J17	0	625	229	156	start	M05
56	T038	J17	0	625	229	155	start	M05
56	T039	J17	0	625	229	166	start	M06
56	T040	J17	0	625	229	155	start	M06
56	T041	J17	0	625	229	154	start	M06
68	T003	J03	0	6250	1244	65	finish	M03
70	T042	J18	6	625	187	15	submit	-
70	T043	J19	6	6250	622	113	submit	-
70	T042	J18	6	625	187	15	start	M03
70	T043	J19	6	6250	622	113	start	M08
77	T004	J04	9	6250	1590	71	finish	M04
85	T042	J18	6	625	187	15	finish	M03
102	T016	J13	6	937	229	62	finish	M07

time	task	job	priority	CPU	memory	run time	event	machine
108	T019	J13	6	937	229	68	finish	M07
109	T017	J13	6	937	229	69	finish	M07
110	T018	J13	6	937	229	70	finish	M07
115	T025	J16	6	937	197	60	finish	M02
118	T023	J16	6	937	197	63	finish	M02
119	T024	J16	6	937	197	64	finish	M02
130	T021	J15	6	645	197	75	finish	M02
130	T022	J16	6	937	197	75	finish	M02
142	T008	J08	6	625	280	120	finish	M05
171	T044	J20	6	625	3180	24	submit	-
171	T044	J20	6	625	3180	24	start	M02
183	T043	J19	6	6250	622	113	finish	M08
195	T044	J20	6	625	3180	24	finish	M02
210	T035	J17	0	625	229	154	finish	M01
210	T041	J17	0	625	229	154	finish	M06
211	T029	J17	0	625	229	155	finish	M01
211	T036	J17	0	625	229	155	finish	M05
211	T038	J17	0	625	229	155	finish	M05
211	T040	J17	0	625	229	155	finish	M06
212	T027	J17	0	625	229	156	finish	M01
212	T030	J17	0	625	229	156	finish	M01
212	T031	J17	0	625	229	156	finish	M01
212	T032	J17	0	625	229	156	finish	M01
212	T033	J17	0	625	229	156	finish	M01
212	T037	J17	0	625	229	156	finish	M05
213	T034	J17	0	625	229	157	finish	M01
222	T013	J12	6	1250	586	182	finish	M06
222	T020	J14	1	1250	466	168	finish	M02
222	T026	J17	0	625	229	166	finish	M01
222	T028	J17	0	625	229	166	finish	M01
222	T039	J17	0	625	229	166	finish	M06
222	T045	J21	1	1250	777	1057	submit	-
222	T045	J21	1	1250	777	1057	start	M01
230	T015	J12	6	1250	586	190	finish	M07
241	T046	J22	0	3125	466	310	submit	-
241	T046	J22	0	3125	466	310	start	M01
251	T014	J12	6	1250	586	211	finish	M06
252	T011	J11	4	625	311	230	finish	M05
255	T047	J23	6	937	1244	298	submit	-
255	T047	J23	6	937	1244	298	start	M01
256	T048	J24	6	937	239	61	submit	-
256	T049	J24	6	937	239	68	submit	-
256	T050	J24	6	937	239	141	submit	-
256	T051	J24	6	937	239	92	submit	-
256	T048	J24	6	937	239	61	start	M01
256	T049	J24	6	937	239	68	start	M02
256	T050	J24	6	937	239	141	start	M02
256	T051	J24	6	937	239	92	start	M02
264	T012	J12	6	1250	586	224	finish	M05
272	T052	J25	0	1688	295	996	submit	-
272	T052	J25	0	1688	295	996	start	M02
274	T006	J06	6	625	280	266	finish	M05
280	T053	J26	6	625	280	157	submit	-
280	T053	J26	6	625	280	157	start	M02
301	T054	J27	0	1250	16	18	submit	-
301	T054	J27	0	1250	16	18	start	M03
309	T055	J28	6	625	187	16	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
309	T056	J29	6	6250	622	132	submit	-
309	T057	J30	6	937	1244	110	submit	-
309	T058	J31	4	3125	448	20	submit	-
309	T055	J28	6	625	187	16	start	M02
309	T056	J29	6	6250	622	132	start	M04
309	T057	J30	6	937	1244	110	start	M03
309	T058	J31	4	3125	448	20	start	M03
317	T048	J24	6	937	239	61	finish	M01
319	T054	J27	0	1250	16	18	finish	M03
324	T049	J24	6	937	239	68	finish	M02
325	T055	J28	6	625	187	16	finish	M02
326	T059	J32	6	634	197	94	submit	-
326	T060	J33	6	937	197	56	submit	-
326	T061	J33	6	937	197	48	submit	-
326	T062	J33	6	937	197	49	submit	-
326	T063	J33	6	937	197	75	submit	-
326	T059	J32	6	634	197	94	start	M01
326	T060	J33	6	937	197	56	start	M02
326	T061	J33	6	937	197	48	start	M02
326	T062	J33	6	937	197	49	start	M03
326	T063	J33	6	937	197	75	start	M03
329	T058	J31	4	3125	448	20	finish	M03
338	T005	J05	1	1250	1590	393	fail	M05
338	T005	J05	1	1250	1590	393	start	M03
348	T051	J24	6	937	239	92	finish	M02
350	T064	J34	2	625	31	83	submit	-
350	T065	J35	2	625	31	28	submit	-
350	T064	J34	2	625	31	83	start	M02
350	T065	J35	2	625	31	28	start	M03
362	T066	J36	6	625	187	18	submit	-
362	T066	J36	6	625	187	18	start	M03
374	T061	J33	6	937	197	48	finish	M02
375	T062	J33	6	937	197	49	finish	M03
378	T065	J35	2	625	31	28	finish	M03
380	T066	J36	6	625	187	18	finish	M03
382	T060	J33	6	937	197	56	finish	M02
397	T050	J24	6	937	239	141	finish	M02
401	T063	J33	6	937	197	75	finish	M03
419	T057	J30	6	937	1244	110	finish	M03
420	T059	J32	6	634	197	94	finish	M01
421	T067	J37	4	3448	1460	599	submit	-
421	T068	J37	4	3448	1460	561	submit	-
421	T069	J37	4	3448	1460	521	submit	-
421	T070	J37	4	3448	1460	595	submit	-
421	T067	J37	4	3448	1460	599	start	M03
421	T068	J37	4	3448	1460	561	start	M06
421	T069	J37	4	3448	1460	521	start	M07
421	T070	J37	4	3448	1460	595	start	M08
433	T064	J34	2	625	31	83	finish	M02
436	T071	J38	4	3125	1224	42	submit	-
436	T071	J38	4	3125	1224	42	start	M02
437	T053	J26	6	625	280	157	finish	M02
441	T056	J29	6	6250	622	132	finish	M04
444	T072	J39	1	1250	777	1094	submit	-
444	T072	J39	1	1250	777	1094	start	M02
448	T073	J40	6	625	280	229	submit	-
448	T074	J41	6	625	3180	23	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
448	T073	J40	6	625	280	229	start	M01
448	T074	J41	6	625	3180	23	start	M03
471	T074	J41	6	625	3180	23	finish	M03
478	T071	J38	4	3125	1224	42	finish	M02
478	T075	J42	6	937	1244	184	submit	-
478	T075	J42	6	937	1244	184	start	M02
480	T076	J43	6	1250	631	184	submit	-
480	T077	J43	6	1250	631	178	submit	-
480	T078	J43	6	1250	631	163	submit	-
480	T079	J43	6	1250	631	183	submit	-
480	T076	J43	6	1250	631	184	start	M02
480	T077	J43	6	1250	631	178	start	M03
480	T078	J43	6	1250	631	163	start	M04
480	T079	J43	6	1250	631	183	start	M04
481	T080	J44	0	6250	1244	62	submit	-
482	T081	J45	0	6250	1244	83	submit	-
492	T082	J46	0	1688	295	1034	submit	-
492	T082	J46	0	1688	295	1034	start	M06
493	T083	J47	4	3125	1125	26	submit	-
493	T084	J48	6	937	253	50	submit	-
493	T085	J48	6	937	253	45	submit	-
493	T086	J48	6	937	253	44	submit	-
493	T087	J48	6	937	253	169	submit	-
493	T084	J48	6	937	253	50	start	M02
493	T085	J48	6	937	253	45	start	M04
493	T086	J48	6	937	253	44	start	M04
493	T087	J48	6	937	253	169	start	M04
506	T088	J49	8	6250	1590	3241	submit	-
511	T089	J50	1	937	777	75	submit	-
511	T089	J50	1	937	777	75	start	M04
513	T090	J51	0	1250	777	73	submit	-
513	T091	J51	0	1250	777	73	submit	-
513	T092	J51	0	1250	777	71	submit	-
513	T090	J51	0	1250	777	73	start	M07
513	T091	J51	0	1250	777	73	start	M07
513	T092	J51	0	1250	777	71	start	M08
514	T093	J52	0	937	277	54	submit	-
514	T094	J52	0	937	277	53	submit	-
514	T095	J52	0	937	277	54	submit	-
514	T096	J52	0	937	277	53	submit	-
514	T097	J52	0	937	277	64	submit	-
514	T098	J52	0	937	277	63	submit	-
514	T099	J52	0	937	277	67	submit	-
514	T100	J52	0	937	277	53	submit	-
514	T101	J52	0	937	277	63	submit	-
514	T102	J52	0	937	277	51	submit	-
514	T103	J52	0	937	277	54	submit	-
514	T104	J52	0	937	277	57	submit	-
514	T105	J52	0	937	277	55	submit	-
514	T106	J52	0	937	277	55	submit	-
514	T107	J52	0	937	277	51	submit	-
514	T108	J52	0	937	277	51	submit	-
514	T093	J52	0	937	277	54	start	M06
514	T094	J52	0	937	277	53	start	M08
520	T109	J53	4	3125	3180	1894	submit	-
520	T110	J53	4	3125	3180	1800	submit	-
520	T111	J53	4	3125	3180	1761	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
520	T112	J53	4	3125	3180	2562	submit	-
535	T113	J54	6	625	187	22	submit	-
536	T114	J55	6	6250	622	121	submit	-
536	T115	J56	6	625	280	45	submit	-
537	T086	J48	6	937	253	44	finish	M04
537	T113	J54	6	625	187	22	start	M04
538	T085	J48	6	937	253	45	finish	M04
538	T115	J56	6	625	280	45	start	M04
543	T084	J48	6	937	253	50	finish	M02
543	T095	J52	0	937	277	54	start	M02
551	T046	J22	0	3125	466	310	finish	M01
551	T096	J52	0	937	277	53	start	M01
551	T097	J52	0	937	277	64	start	M01
553	T047	J23	6	937	1244	298	finish	M01
553	T098	J52	0	937	277	63	start	M01
553	T099	J52	0	937	277	67	start	M01
559	T113	J54	6	625	187	22	finish	M04
559	T100	J52	0	937	277	53	start	M04
567	T094	J52	0	937	277	53	finish	M08
567	T101	J52	0	937	277	63	start	M08
568	T093	J52	0	937	277	54	finish	M06
568	T102	J52	0	937	277	51	start	M06
581	T116	J57	6	625	280	118	submit	-
581	T116	J57	6	625	280	118	start	M01
583	T115	J56	6	625	280	45	finish	M04
583	T103	J52	0	937	277	54	start	M04
584	T092	J51	0	1250	777	71	finish	M08
584	T104	J52	0	937	277	57	start	M08
586	T089	J50	1	937	777	75	finish	M04
586	T090	J51	0	1250	777	73	finish	M07
586	T091	J51	0	1250	777	73	finish	M07
586	T105	J52	0	937	277	55	start	M04
586	T106	J52	0	937	277	55	start	M07
586	T107	J52	0	937	277	51	start	M07
597	T095	J52	0	937	277	54	finish	M02
597	T108	J52	0	937	277	51	start	M02
601	T117	J58	9	6250	233	115	submit	-
603	T118	J59	0	6250	466	34	submit	-
604	T096	J52	0	937	277	53	finish	M01
612	T100	J52	0	937	277	53	finish	M04
612	T119	J60	8	625	466	1395	submit	-
612	T120	J61	4	1250	398	19	submit	-
612	T121	J62	4	625	311	13	submit	-
612	T119	J60	8	625	466	1395	start	M01
612	T121	J62	4	625	311	13	start	M04
613	T122	J63	6	937	197	58	submit	-
615	T097	J52	0	937	277	64	finish	M01
615	T122	J63	6	937	197	58	start	M01
616	T098	J52	0	937	277	63	finish	M01
616	T123	J64	9	6250	1590	80	submit	-
616	T120	J61	4	1250	398	19	start	M01
619	T102	J52	0	937	277	51	finish	M06
620	T099	J52	0	937	277	67	finish	M01
625	T121	J62	4	625	311	13	finish	M04
628	T124	J65	4	3125	1125	66	submit	-
630	T101	J52	0	937	277	63	finish	M08
635	T120	J61	4	1250	398	19	finish	M01

time	task	job	priority	CPU	memory	run time	event	machine
637	T103	J52	0	937	277	54	finish	M04
637	T107	J52	0	937	277	51	finish	M07
641	T104	J52	0	937	277	57	finish	M08
641	T105	J52	0	937	277	55	finish	M04
641	T106	J52	0	937	277	55	finish	M07
643	T078	J43	6	1250	631	163	finish	M04
648	T108	J52	0	937	277	51	finish	M02
658	T077	J43	6	1250	631	178	finish	M03
662	T075	J42	6	937	1244	184	finish	M02
662	T087	J48	6	937	253	169	finish	M04
663	T079	J43	6	1250	631	183	finish	M04
663	T117	J58	9	6250	233	115	start	M04
664	T076	J43	6	1250	631	184	finish	M02
664	T052	J25	0	1688	295	996	evict	M02
664	T072	J39	1	1250	777	1094	evict	M02
664	T123	J64	9	6250	1590	80	start	M02
664	T072	J39	1	1250	777	1094	start	M01
664	T052	J25	0	1688	295	996	start	M07
668	T125	J66	6	625	280	223	submit	-
668	T125	J66	6	625	280	223	start	M01
669	T126	J67	4	937	549	806	submit	-
669	T127	J67	4	937	549	809	submit	-
669	T128	J67	4	937	549	797	submit	-
669	T129	J67	4	937	549	807	submit	-
669	T126	J67	4	937	549	806	start	M03
669	T127	J67	4	937	549	809	start	M06
669	T128	J67	4	937	549	797	start	M07
669	T129	J67	4	937	549	807	start	M08
673	T122	J63	6	937	197	58	finish	M01
677	T073	J40	6	625	280	229	finish	M01
695	T130	J68	6	937	1244	185	submit	-
695	T131	J69	6	625	280	107	submit	-
695	T131	J69	6	625	280	107	start	M01
697	T132	J70	6	1250	637	184	submit	-
697	T133	J70	6	1250	637	164	submit	-
697	T134	J70	6	1250	637	181	submit	-
697	T135	J70	6	1250	637	166	submit	-
697	T132	J70	6	1250	637	184	start	M08
699	T116	J57	6	625	280	118	finish	M01
710	T136	J71	6	625	3180	27	submit	-
725	T137	J72	6	937	1244	68	submit	-
726	T138	J73	6	634	197	68	submit	-
726	T139	J73	6	634	197	66	submit	-
726	T140	J74	6	937	197	66	submit	-
726	T141	J74	6	937	197	68	submit	-
726	T142	J74	6	937	197	64	submit	-
726	T143	J74	6	937	197	63	submit	-
726	T138	J73	6	634	197	68	start	M01
726	T139	J73	6	634	197	66	start	M01
731	T005	J05	1	1250	1590	393	finish	M03
731	T130	J68	6	937	1244	185	start	M03
744	T123	J64	9	6250	1590	80	finish	M02
744	T088	J49	8	6250	1590	3241	start	M02
778	T117	J58	9	6250	233	115	finish	M04
778	T114	J55	6	6250	622	121	start	M04
788	T144	J75	6	6250	622	115	submit	-
788	T145	J76	6	625	280	55	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
788	T145	J76	6	625	280	55	start	M03
792	T139	J73	6	634	197	66	finish	M01
794	T138	J73	6	634	197	68	finish	M01
794	T146	J77	1	937	622	236	submit	-
796	T147	J78	0	1250	622	130	submit	-
796	T148	J78	0	1250	622	116	submit	-
796	T149	J78	0	1250	622	115	submit	-
796	T150	J78	0	1250	622	141	submit	-
796	T151	J78	0	1250	622	195	submit	-
796	T152	J78	0	1250	622	139	submit	-
796	T153	J78	0	1250	622	162	submit	-
796	T154	J78	0	1250	622	176	submit	-
796	T155	J78	0	1250	622	117	submit	-
796	T156	J78	0	1250	622	119	submit	-
796	T157	J78	0	1250	622	157	submit	-
796	T158	J78	0	1250	622	119	submit	-
796	T159	J78	0	1250	622	170	submit	-
796	T160	J78	0	1250	622	119	submit	-
796	T161	J78	0	1250	622	180	submit	-
796	T162	J78	0	1250	622	126	submit	-
802	T131	J69	6	625	280	107	finish	M01
802	T133	J70	6	1250	637	164	start	M01
820	T163	J79	6	625	3180	31	submit	-
843	T145	J76	6	625	280	55	finish	M03
854	T164	J80	6	625	187	20	submit	-
854	T165	J81	4	62	398	17	submit	-
854	T164	J80	6	625	187	20	start	M03
854	T165	J81	4	62	398	17	start	M03
871	T165	J81	4	62	398	17	finish	M03
874	T164	J80	6	625	187	20	finish	M03
881	T132	J70	6	1250	637	184	finish	M08
881	T134	J70	6	1250	637	181	start	M08
891	T125	J66	6	625	280	223	finish	M01
891	T140	J74	6	937	197	66	start	M01
891	T141	J74	6	937	197	68	start	M01
899	T114	J55	6	6250	622	121	finish	M04
899	T135	J70	6	1250	637	166	start	M04
899	T136	J71	6	625	3180	27	start	M04
899	T137	J72	6	937	1244	68	start	M04
899	T142	J74	6	937	197	64	start	M04
899	T143	J74	6	937	197	63	start	M04
899	T147	J78	0	1250	622	130	start	M04
903	T166	J82	6	625	280	129	submit	-
903	T166	J82	6	625	280	129	start	M03
916	T130	J68	6	937	1244	185	finish	M03
916	T163	J79	6	625	3180	31	start	M03
926	T136	J71	6	625	3180	27	finish	M04
942	T069	J37	4	3448	1460	521	finish	M07
942	T167	J83	6	645	197	74	submit	-
942	T168	J83	6	645	197	76	submit	-
942	T169	J84	6	937	197	74	submit	-
942	T170	J84	6	937	197	73	submit	-
942	T171	J84	6	937	197	74	submit	-
942	T172	J84	6	937	197	74	submit	-
942	T167	J83	6	645	197	74	start	M04
942	T168	J83	6	645	197	76	start	M07
942	T169	J84	6	937	197	74	start	M07

time	task	job	priority	CPU	memory	run time	event	machine
942	T170	J84	6	937	197	73	start	M07
942	T171	J84	6	937	197	74	start	M07
947	T163	J79	6	625	3180	31	finish	M03
947	T172	J84	6	937	197	74	start	M03
957	T140	J74	6	937	197	66	finish	M01
959	T141	J74	6	937	197	68	finish	M01
962	T143	J74	6	937	197	63	finish	M04
963	T142	J74	6	937	197	64	finish	M04
963	T173	J85	0	6250	1244	53	submit	-
966	T133	J70	6	1250	637	164	finish	M01
966	T174	J86	0	6250	1244	51	submit	-
966	T146	J77	1	937	622	236	start	M01
967	T137	J72	6	937	1244	68	finish	M04
982	T068	J37	4	3448	1460	561	finish	M06
982	T083	J47	4	3125	1125	26	start	M06
989	T175	J87	6	937	1244	200	submit	-
990	T176	J88	6	1250	633	153	submit	-
990	T177	J88	6	1250	633	165	submit	-
990	T178	J88	6	1250	633	153	submit	-
990	T179	J88	6	1250	633	171	submit	-
990	T180	J89	6	937	258	199	submit	-
990	T181	J89	6	937	258	113	submit	-
990	T182	J89	6	937	258	46	submit	-
990	T183	J89	6	937	258	42	submit	-
990	T180	J89	6	937	258	199	start	M01
990	T182	J89	6	937	258	46	start	M04
990	T183	J89	6	937	258	42	start	M04
1008	T083	J47	4	3125	1125	26	finish	M06
1008	T175	J87	6	937	1244	200	start	M06
1008	T176	J88	6	1250	633	153	start	M06
1008	T177	J88	6	1250	633	165	start	M06
1015	T170	J84	6	937	197	73	finish	M07
1015	T184	J90	1	937	777	86	submit	-
1015	T181	J89	6	937	258	113	start	M07
1016	T070	J37	4	3448	1460	595	finish	M08
1016	T167	J83	6	645	197	74	finish	M04
1016	T169	J84	6	937	197	74	finish	M07
1016	T171	J84	6	937	197	74	finish	M07
1016	T178	J88	6	1250	633	153	start	M07
1016	T179	J88	6	1250	633	171	start	M08
1016	T184	J90	1	937	777	86	start	M08
1017	T185	J91	0	1250	777	83	submit	-
1017	T186	J91	0	1250	777	80	submit	-
1017	T187	J91	0	1250	777	82	submit	-
1017	T188	J91	0	1250	777	84	submit	-
1017	T189	J91	0	1250	777	83	submit	-
1017	T190	J91	0	1250	777	74	submit	-
1017	T191	J91	0	1250	777	73	submit	-
1017	T192	J91	0	1250	777	82	submit	-
1017	T193	J91	0	1250	777	81	submit	-
1017	T194	J91	0	1250	777	71	submit	-
1018	T168	J83	6	645	197	76	finish	M07
1018	T148	J78	0	1250	622	116	start	M07
1019	T195	J92	0	937	277	58	submit	-
1019	T196	J92	0	937	277	49	submit	-
1019	T197	J92	0	937	277	50	submit	-
1019	T198	J92	0	937	277	48	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
1019	T199	J92	0	937	277	47	submit	-
1019	T200	J92	0	937	277	49	submit	-
1019	T201	J92	0	937	277	74	submit	-
1019	T202	J92	0	937	277	58	submit	-
1019	T203	J92	0	937	277	52	submit	-
1019	T204	J92	0	937	277	47	submit	-
1019	T205	J92	0	937	277	51	submit	-
1019	T206	J92	0	937	277	63	submit	-
1019	T207	J92	0	937	277	47	submit	-
1019	T208	J92	0	937	277	46	submit	-
1019	T209	J92	0	937	277	48	submit	-
1019	T210	J92	0	937	277	47	submit	-
1019	T195	J92	0	937	277	58	start	M08
1019	T208	J92	0	937	277	46	start	M04
1020	T067	J37	4	3448	1460	599	finish	M03
1020	T109	J53	4	3125	3180	1894	start	M03
1021	T172	J84	6	937	197	74	finish	M03
1021	T149	J78	0	1250	622	115	start	M03
1029	T147	J78	0	1250	622	130	finish	M04
1032	T166	J82	6	625	280	129	finish	M03
1032	T183	J89	6	937	258	42	finish	M04
1032	T196	J92	0	937	277	49	start	M03
1036	T182	J89	6	937	258	46	finish	M04
1041	T211	J93	4	6250	2188	16	submit	-
1062	T134	J70	6	1250	637	181	finish	M08
1062	T150	J78	0	1250	622	141	start	M08
1065	T135	J70	6	1250	637	166	finish	M04
1065	T208	J92	0	937	277	46	finish	M04
1065	T144	J75	6	6250	622	115	start	M04
1077	T195	J92	0	937	277	58	finish	M08
1077	T197	J92	0	937	277	50	start	M08
1081	T196	J92	0	937	277	49	finish	M03
1081	T198	J92	0	937	277	48	start	M03
1102	T184	J90	1	937	777	86	finish	M08
1102	T151	J78	0	1250	622	195	start	M08
1127	T197	J92	0	937	277	50	finish	M08
1127	T152	J78	0	1250	622	139	start	M08
1128	T181	J89	6	937	258	113	finish	M07
1129	T198	J92	0	937	277	48	finish	M03
1129	T199	J92	0	937	277	47	start	M03
1134	T148	J78	0	1250	622	116	finish	M07
1136	T149	J78	0	1250	622	115	finish	M03
1136	T153	J78	0	1250	622	162	start	M03
1142	T212	J94	0	3125	466	266	submit	-
1161	T176	J88	6	1250	633	153	finish	M06
1161	T154	J78	0	1250	622	176	start	M06
1169	T178	J88	6	1250	633	153	finish	M07
1169	T110	J53	4	3125	3180	1800	start	M07
1173	T177	J88	6	1250	633	165	finish	M06
1173	T155	J78	0	1250	622	117	start	M06
1176	T199	J92	0	937	277	47	finish	M03
1176	T200	J92	0	937	277	49	start	M03
1180	T144	J75	6	6250	622	115	finish	M04
1180	T111	J53	4	3125	3180	1761	start	M04
1180	T124	J65	4	3125	1125	66	start	M04
1187	T179	J88	6	1250	633	171	finish	M08
1187	T156	J78	0	1250	622	119	start	M08

time	task	job	priority	CPU	memory	run time	event	machine
1189	T180	J89	6	937	258	199	finish	M01
1189	T201	J92	0	937	277	74	start	M01

Cell window 24: Events

Time 338: machine M05 failed; task T005 stopped and went back to the queue.

Time 338: task T005 restarted on machine M03.

Time 664: task T123 (priority 9) evicted task T052 (priority 0) from machine M02.

Time 664: task T123 (priority 9) evicted task T072 (priority 1) from machine M02.

Time 664: task T052 restarted on machine M07.

Time 664: task T072 restarted on machine M01.

Longest waits (submission to first start, tasks started by the window end): task T111 waited 660 s (submitted at 520, first started at 1180); task T110 waited 649 s (submitted at 520, first started at 1169); task T124 waited 552 s (submitted at 628, first started at 1180); task T109 waited 500 s (submitted at 520, first started at 1020); task T083 waited 489 s (submitted at 493, first started at 982).

Cell window 24: Queue at window end

State after the events at time 1200.

Tasks running at time 1200 (remaining run time = seconds still to run after time 1200):

task	job	priority	CPU	memory	machine	started	remaining
T045	J21	1	1250	777	M01	222	79
T072	J39	1	1250	777	M01	664	558
T119	J60	8	625	466	M01	612	807
T146	J77	1	937	622	M01	966	2
T201	J92	0	937	277	M01	1189	63
T088	J49	8	6250	1590	M02	744	2785
T109	J53	4	3125	3180	M03	1020	1714
T126	J67	4	937	549	M03	669	275
T153	J78	0	1250	622	M03	1136	98
T200	J92	0	937	277	M03	1176	25
T111	J53	4	3125	3180	M04	1180	1741
T124	J65	4	3125	1125	M04	1180	46
T082	J46	0	1688	295	M06	492	326
T127	J67	4	937	549	M06	669	278
T154	J78	0	1250	622	M06	1161	137
T155	J78	0	1250	622	M06	1173	90
T175	J87	6	937	1244	M06	1008	8
T052	J25	0	1688	295	M07	664	460
T110	J53	4	3125	3180	M07	1169	1769
T128	J67	4	937	549	M07	669	266
T129	J67	4	937	549	M08	669	276
T150	J78	0	1250	622	M08	1062	3
T151	J78	0	1250	622	M08	1102	97
T152	J78	0	1250	622	M08	1127	66
T156	J78	0	1250	622	M08	1187	106

Tasks waiting at time 1200, in queue order (each will run its full run time once started):

task	job	priority	CPU	memory	submitted	run time
T112	J53	4	3125	3180	520	2562
T211	J93	4	6250	2188	1041	16
T080	J44	0	6250	1244	481	62

task	job	priority	CPU	memory	submitted	run time
T081	J45	0	6250	1244	482	83
T118	J59	0	6250	466	603	34
T157	J78	0	1250	622	796	157
T158	J78	0	1250	622	796	119
T159	J78	0	1250	622	796	170
T160	J78	0	1250	622	796	119
T161	J78	0	1250	622	796	180
T162	J78	0	1250	622	796	126
T173	J85	0	6250	1244	963	53
T174	J86	0	6250	1244	966	51
T185	J91	0	1250	777	1017	83
T186	J91	0	1250	777	1017	80
T187	J91	0	1250	777	1017	82
T188	J91	0	1250	777	1017	84
T189	J91	0	1250	777	1017	83
T190	J91	0	1250	777	1017	74
T191	J91	0	1250	777	1017	73
T192	J91	0	1250	777	1017	82
T193	J91	0	1250	777	1017	81
T194	J91	0	1250	777	1017	71
T202	J92	0	937	277	1019	58
T203	J92	0	937	277	1019	52
T204	J92	0	937	277	1019	47
T205	J92	0	937	277	1019	51
T206	J92	0	937	277	1019	63
T207	J92	0	937	277	1019	47
T209	J92	0	937	277	1019	48
T210	J92	0	937	277	1019	47
T212	J94	0	3125	466	1142	266

Machine M05 is down at time 1200; it returns at time 11189.